

Faculty of Applied Sciences

Master's Thesis

Part-Load Performance Analysis of a Transcritical CO_2 Low-Temperature Power Cycle

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Abstract

Carnot batteries represent a promising solution for grid-scale energy storage, with transcritical CO_2 Rankine cycles standing out due to compact equipment, near-ambient condensation, and favorable temperature-glide matching. However, the non-linear thermophysical behavior of CO_2 near the critical point makes these cycles highly sensitive to boundary conditions, leaving their control relatively understudied compared to heat pumps.

Regarding the on-design analysis, a steady-state model was implemented in Python using *LaboThApPy*, employing discretized heat exchangers to capture non-linear c_p variations. A Particle Swarm Optimization algorithm identified the operating points maximizing net cycle efficiency for a fixed 10 MW constraint across hot-source (130–200 °C) and cold-source (0.1–20 °C) temperature maps. The global optimum achieved 20.3 % net efficiency at 218 bar. Across the explored envelope, CO_2 and hot-source mass flow rates varied by factors of three and four, respectively, quantifying the high sensitivity of the cycle to its boundary conditions and identifying the high-side pressure importance as controlled variable.

For the off-design analysis, the component performances were no longer prescribed: the hardware was fixed at the geometry resulting from the design point, the heat exchangers were modelled with a moving-boundary method and the turbomachinery with performance maps, while the cycle was closed through a fixed sub-cooling target. The pump and expander rotational speeds (N_{pp}, N_{exp}) were swept around their nominal values, and the analysis was repeated under independent variations of the three boundary conditions (T_{hf}, T_{cf}, $\dot{m}_{hf}$). The results show that the high-side pressure and the CO_2 mass flow rate are governed almost exclusively by the pump speed and remain essentially insensitive to the thermal boundary conditions; that the cold-source temperature is the principal disturbance, acting directly on the pump-suction density; and that the hot-source mass flow rate behaves as an energy regulator with clearly diminishing returns. The expander speed was found to be beneficial only when the pump already sustains sufficient flow.

On this basis, a variable structure for the power block is proposed in which the high-side pressure emerges as the central controlled variable, regulated mainly through the pump speed, the cold-source temperature is treated as the main measured disturbance to be fed forward, and the hot-source mass flow rate and the expander speed act as secondary actuators whose primary effect is on cycle efficiency and net power output rather than on pressure regulation. These results provide reference operating trajectories and a justified set of manipulated and controlled variables that constitute a set of guidelines for the design and tuning of future control strategies for transcritical CO_2 power cycles in Carnot battery applications.

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1 Introduction

Nowadays, the global energy landscape is going through a transition towards renewable energy sources (RES). However, one of the main disadvantages of these sources is their intermittent nature which creates a mismatch between power production and demand. To ensure grid stability and prevent the limitation of excessive renewable electricity, a massive expansion of energy storage is required. While approximately 235 GW of energy storage was installed globally by the end of 2023, the International Energy Agency (IEA) projects that 1500 GW of total grid-scale storage capacity will be needed by 2030 to reach the Net Zero Scenario, of which battery storage is expected to account for approximately 90 % [1]. This gap highlights the need to store renewable energy for later grid use, which is one of the primary purposes of a Carnot Battery. Basically, a Carnot Battery converts electricity into heat using a heat pump or an electric heating element, and then converts the stored heat back into electricity using thermal engines as required.

At the University of Liège (ULiège), the objective is to develop a Carnot Battery which will use transcritical Rankine cycle technology using CO_2 as a working fluid, taking advantage of the interesting properties of this substance. A CO_2 Carnot Battery can achieve high cycle efficiency and compact equipment [2]. However, while the transcritical CO_2 power cycle has been studied in the literature, its part-load and off-design operation remains comparatively underexplored.

The operation of transcritical CO_2 cycles near the critical point introduces strong non-linear behaviour, where small variations in pressure or temperature can significantly affect efficiency and operational stability. In particular, variables such as high-side pressure, turbine inlet temperature, mass flow rate and working-fluid inventory are strongly coupled, making the identification of suitable operating strategies challenging. Although some studies have investigated pressure and inventory management in CO_2 power systems [3], the control of transcritical Rankine configurations remains insufficiently explored.Therefore, the objective of this thesis is to perform a control-oriented thermodynamic analysis of a transcritical CO_2 Rankine cycle for Carnot Battery applications. Using a steady-state model, the work aims to identify the key operating parameters influencing cycle performance under on-design and off-design conditions. Particular attention is given to the influence of pressure levels, source temperatures, mass flow rates and turbomachinery operation on cycle efficiency and power output.

As a first step toward future control development, the thesis proposes and evaluates a set of manipulated and controlled variables for the transcritical CO_2 power block. The sensitivity analysis presented in the following chapters provides the basis for future control strategies and off-design operation studies.

2 State Of the Art

The aim of this chapter is to provide the necessary context for understanding the research presented in this thesis. It begins with an overview of the thermodynamic and engineering principles behind Carnot batteries and gradually focuses on the specific case under study: a transcritical CO_2 cycle in this context. Along the way, it surveys the existing literature on transcritical CO_2 power cycles, control strategies for related thermodynamic systems, and Carnot battery configurations, in order to identify what has already been investigated and where the key gaps remain. In this way, the chapter can be seen as a guided progression from a general framework to the specific problem addressed in this thesis.

2.1 Carnot Batteries

2.1.1 Concept and thermodynamic principles

Carnot batteries are a type of long-duration energy storage that converts electricity to heat (charge phase) and back to electricity (discharge phase). In charging mode, surplus electric power from RES (e.g. from wind or solar during off-peak periods) is used to run a heat pump or resistance heater, producing high-temperature heat that is stored in a thermal medium such as molten salt, rocks, water or other materials. In discharge mode, this stored heat drives a heat engine to generate electricity when needed. In short, a Carnot battery follows the path : electricity $\rightarrow$ heat $\rightarrow$ electricity.

As illustrated in Figure 1, this process relies on four main interacting domains: a heat pump (HP), a heat engine (HE) and two thermal reservoirs (High T and Low T). During the charge phase, the HP consumes electrical energy (E_{hp}) to extract heat from the low temperature reservoir, transferring high-temperature thermal energy ($Q_{HT,hp}$) into the high temperature storage. Conversely, during the discharge phase, the HE absorbs this stored heat ($Q_{HT,he}$) to generate useful electrical power (E_{he}), while rejecting the remaining low-grade heat ($Q_{LT,he}$) back to the low temperature reservoir. Furthermore, this schematic highlights the potential for external thermal integration, allowing external heat flows ($Q_{HT,ext}$) to supplement the system and improve its overall performance.

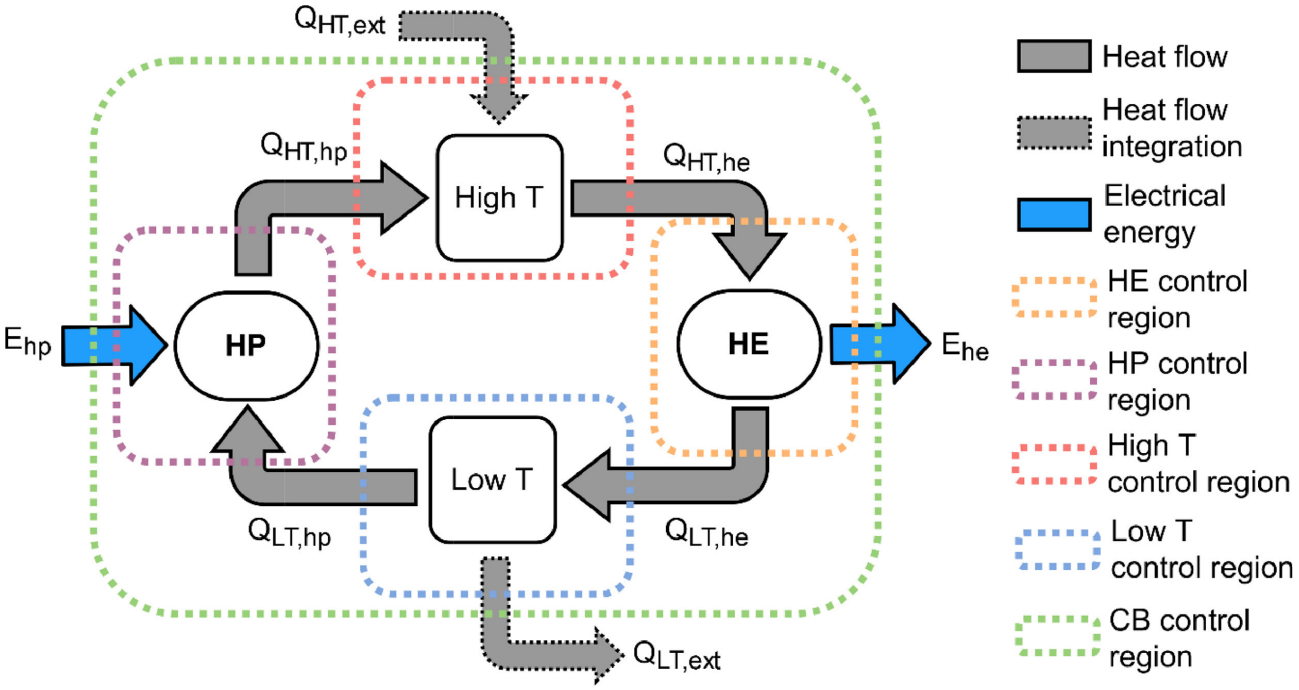


Figure 1: Schematic of a Carnot Battery system showing the charge and discharge phases. Adapted from [4]

The key performance metric is the Round-Trip Efficiency (RTE), defined as the ratio of electrical energy output during discharge to electrical energy input during charging:

$$RTE = \frac{E_{discharge}}{E_{charge}} \tag{1}$$

where $E_{discharge}$ is the net electrical output from the heat engine and E_{charge} is the electrical input to the heat pump. For a standard Carnot Battery without waste-heat integration, RTE is typically below 70 % [5, 6]. When waste heat is thermally integrated, some studies report RTEs exceeding 100 %, because additional exergy from an external thermal source supplements the stored electrical energy [7].

2.1.2 Advantages

Carnot Batteries present several advantages compared to other grid-scale energy storage technologies:

- **Sustainability:** They rely on abundant and non-toxic materials (e.g. water, molten salts, rocks), avoiding critical raw materials such as lithium or cobalt [8].
- **Geographic flexibility:** Unlike pumped hydro storage, they do not require specific geographical conditions and can be deployed in a wide range of locations.

- **Sector Coupling:** Since energy is stored thermally, Carnot Batteries can simultaneously deliver electricity and useful heat (e.g. for district heating), increasing overall system efficiency.
- **Low cost for long-duration storage:** The use of inexpensive thermal storage media could make Carnot Batteries cost-competitive with electrochemical alternatives, particularly for long-duration applications where the levelized cost of storage (LCOS) is more sensitive to energy capacity costs than to power conversion costs [5].
- **Long lifetime:** Carnot battery systems are expected to last decades. Since they do not rely on chemical reactions, there is minimal degradation over time. Electrochemical batteries typically have a lifetime below 10 years, whereas Carnot-based systems could have an expected lifetime of around 20 years [5].

2.1.3 Classification of Carnot Battery Configurations

Carnot Battery systems are generally classified according to the thermodynamic cycle used in the charging and discharging processes. The two main categories are:

- **Brayton cycle Carnot Batteries**
  These systems operate with gas cycles (e.g. air or argon) and enable very high temperature levels (up to 1000 °C), leading to high theoretical efficiencies. However, they require advanced storage materials and turbomachinery due to the extreme operating conditions [9].
- **Rankine cycle Carnot Batteries**
  Rankine systems use a phase-changing working fluid. During charging, a vapour-compression heat pump raises the temperature of the high-temperature reservoir; during discharging, an ORC or steam turbine converts the stored heat back to electricity. Compared to Brayton systems, Rankine CBs operate at much lower temperatures (typically below 200 °C), making them easier to build with commercially available components and more compatible with waste-heat sources. For these reasons and advantages the Rankine cycle is selected in this project[4].

2.1.4 Thermal Energy Storage

All Carnot Batteries require a thermal energy storage (TES) system, which can be broadly classified into three main categories. Sensible heat storage (STES) is the most mature and cost-effective option, although it offers relatively low energy density. Latent heat storage (LTES), based on phase change materials, provides higher energy density but comes with increased cost

and system complexity. Finally, thermochemical storage (TCES) offers the highest energy density and the advantage of negligible self-discharge, although it is still at a relatively low level of technological maturity. [4, 5]

In this project, sensible heat storage has been selected as the thermal energy storage medium for both reservoirs. This choice is justified by reasons aligned with the characteristics of the transcritical CO_2 cycle. First, the high-temperature reservoir of the cycle will operate at moderate temperatures, typically between 70 and 170 ºC, which is well within the range of pressurised water or thermal oils. Second, these fluids are cheap and abundant. Third, the low-temperature reservoir operates under ambient conditions (below 30 ºC) where liquid water is the most practical and cost-effective medium. It is worth noting, however, that ice storage represents a particularly attractive alternative for the cold side of transcritical CO_2 cycles: since CO_2 condenses near ambient temperature, coupling the condenser with an ice storage unit allows the condensation pressure to approach 0 ºC, significantly improving the cycle's pressure ratio and thermodynamic efficiency [6]. Finally, sensible storage is a commercially mature technology with a well-understood cost structure and minimal operational complexity [10].

2.2 Transcritical CO_2 Cycle

2.2.1 The critical point and CO_2 as a working fluid

Every pure substance has a critical point: a specific combination of temperature and pressure above which the distinction between liquid and vapour phase ceases to exist and the fluid enters the supercritical state. As shown in Figure 2, for CO_2 this point lies at $T_c = 31.1\,°C$ and $P_c = 7.38\,MPa$. The main properties that justify the selection of CO_2 as the working fluid in this project are the following:

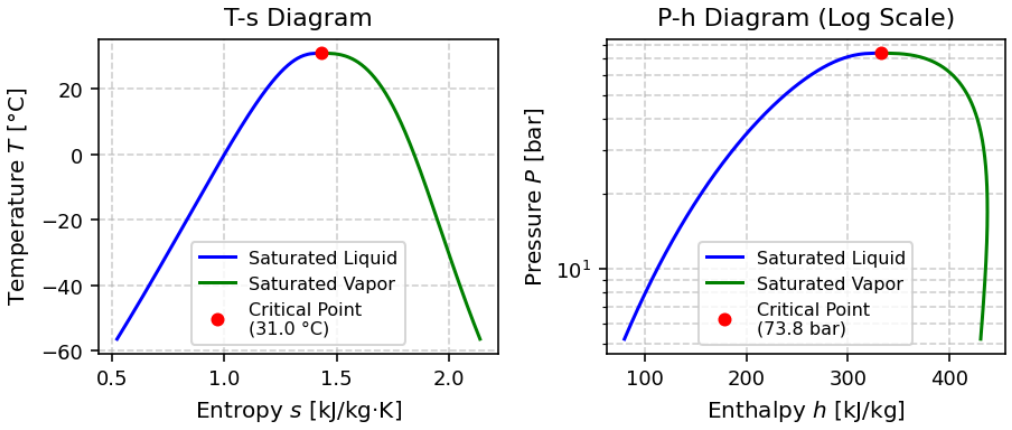


Figure 2: Thermodynamic phase diagrams for CO_2. The critical point marks the apex of the saturation dome, above which the fluid transitions into the supercritical regime.

The main properties that justify the selection of CO_2 as the working fluid in this project are the following:

- **Compactness.** Near the critical point the density of CO_2 is very high, allowing turbomachinery to be made significantly more compact than equivalent steam or conventional ORC machines [2]. Smaller equipment implies lower material costs, which is important for commercial viability.
- **High volumetric heat capacity and non-linear c_p behaviour.** Near the critical point the specific heat c_p of CO_2 increases sharply, reaching very high values that allow heat exchangers to transfer large amounts of thermal energy with small fluid flow rates [11]. As illustrated in Figure 3, this peak is most pronounced at pressures close to the critical pressure ($P_c = 7.38\,\text{MPa}$) and broadens progressively at higher pressures [12]. This strongly non-linear c_p behaviour is both an advantage and a challenge: it enables effective temperature-glide matching with the hot source in the gas heater, reducing entropy generation, but also introduces a non-smooth heating profile that must be carefully accounted for in the heat exchanger modelling and during system operation.

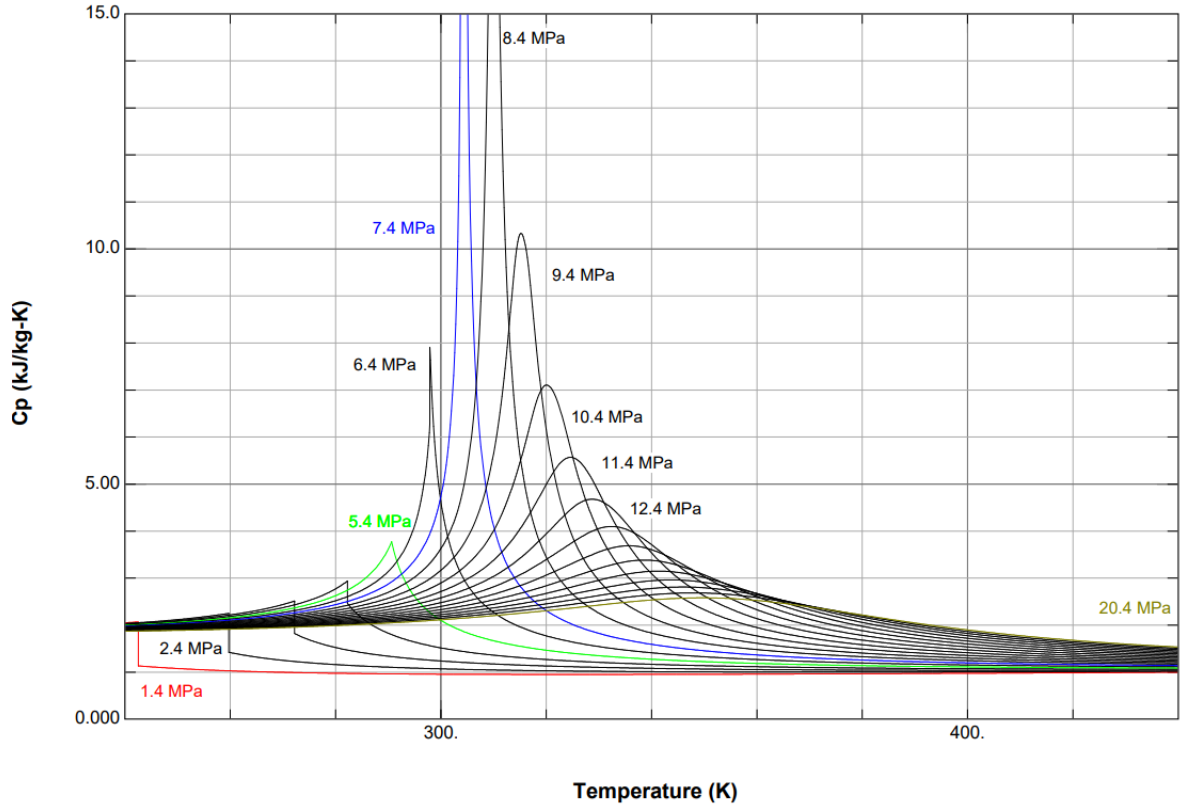


Figure 3: Specific Heat (c_p) vs Temperature at various Pressures for Carbon Dioxide. The blue line indicates the critical pressure of 7.4 MPa. Adapted from [12].

- **Pump advantage over compressor.** In a transcritical cycle the low-pressure side operates in the two-phase region. Condensing the fluid to liquid before pressurisation reduces the required work substantially, since pump work scales with the specific volume of the fluid, which is orders of magnitude smaller for a liquid than for a gas. The transcritical architecture therefore avoids the large compressor work of a purely supercritical Brayton cycle [10].
- **Versatility.** The operating pressure and temperature range of transcritical CO_2 cycles is compatible with a wide variety of heat sources, including industrial waste heat, solar thermal and geothermal applications [10].
- **Safety.** CO_2 is non-flammable, non-toxic at moderate concentrations, non-explosive and non-corrosive towards most common structural materials, making it considerably easier and cheaper to handle than flammable refrigerants such as propane or toxic fluids such as ammonia.
- **Environmental compatibility.** CO_2 has zero ozone depletion potential (ODP = 0) and a global warming potential of 1 (GWP = 1), the baseline against which all other greenhouse gases are measured. Lorentzen [13], who pioneered the revival of CO_2 as a working fluid after decades of displacement by synthetic refrigerants, highlighted precisely these environmental advantages as the primary motivation for returning to it. In a closed thermodynamic loop it does not contribute to atmospheric warming, unlike hydrofluorocarbons.

2.2.2 From subcritical to supercritical: three operating regimes

The operation of CO_2 power cycles can be classified into three regimes based on how the operating pressures relate to the critical point: Subcritical, Transcritical, and Supercritical. Each regime is described below.

Subcritical Cycles
In a conventional subcritical Rankine cycle, the type used in most power plants and ORC systems, the working fluid circulates entirely below its critical point. On the high-pressure side, the fluid evaporates (liquid to vapour) by absorbing heat from a hot source and, on the low-pressure side, the vapour condenses (vapour to liquid) by rejecting heat to a cold source. This two-phase behaviour is thermodynamically convenient because evaporation and condensation occur at constant temperature.

Subcritical cycles are widely implemented in industrial applications. For example, several studies have investigated the application of subcritical ORC systems for industrial waste heat recovery. Quoilin et al. [14] reviewed the use of ORC technology for low and medium-temperature heat sources and highlighted their suitability for decentralized power generation. Zhang et al.

investigated subcritical ORC systems coupled with geothermal sources and reported thermal efficiencies between 8 % and 15 % depending on the heat-source temperature and working fluid selection [15]. These studies demonstrate the technological maturity and operational stability of subcritical configurations, particularly under relatively constant operating conditions. However, they also have limitations: the maximum pressure and temperature are constrained by the saturation curve and the enthalpy difference across the turbine (and therefore the specific work output) is limited, particularly for fluids with low critical temperatures. See Figure 4.

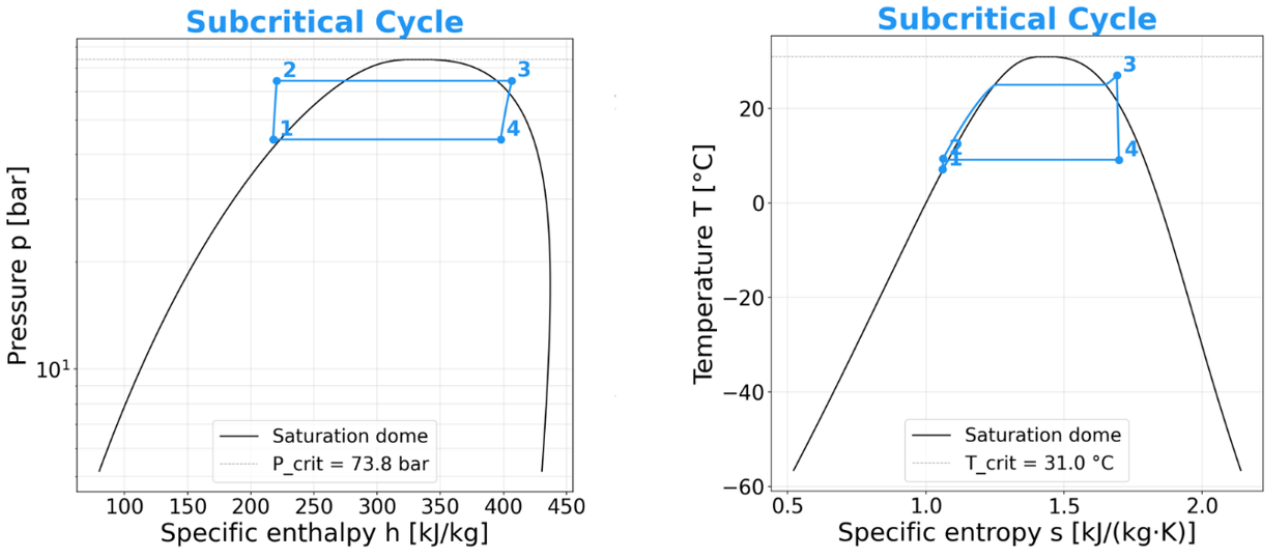


Figure 4: Diagram T-s and P-h of a Subcritical Cycle

Supercritical Cycles

A supercritical cycle, such as the supercritical CO_2 (sCO_2) Brayton cycle that has attracted significant research interest for power generation, operates entirely above the critical point. There is no phase change; the fluid remains in a single, dense supercritical phase throughout. The primary advantage of operating near the critical point is that the fluid density is exceptionally high, which dramatically reduces the compression work (since it is easier to compress a dense fluid than a light gas). Furthermore, this consistently high density significantly reduces the physical size of the turbomachinery and heat exchangers, resulting in a remarkably compact system footprint compared to conventional steam Rankine plants [3].

Supercritical CO_2 Brayton cycles have been extensively studied for high-temperature energy applications. For example, Dostal proposed the recompression sCO_2 Brayton cycle for Generation IV nuclear reactors and demonstrated efficiencies above 45 % at turbine inlet temperatures around 550 °C [16]. Similarly, supercritical CO_2 cycles have been investigated for concentrated

solar power (CSP) applications due to their compactness and reduced cooling requirements compared to steam Rankine systems [17]. These studies largely focus on maximizing thermal efficiency and recuperator effectiveness under high-temperature operating conditions.

However, these systems also present notable disadvantages. Purely supercritical Brayton cycles typically require very high heat source temperatures (between 350 and 600 °C) and high pressures (typically 20–30 MPa) to achieve competitive efficiencies, which pushes the boundaries of available materials, sealing techniques, and turbomachinery technology[16, 17]. See Figure 5.

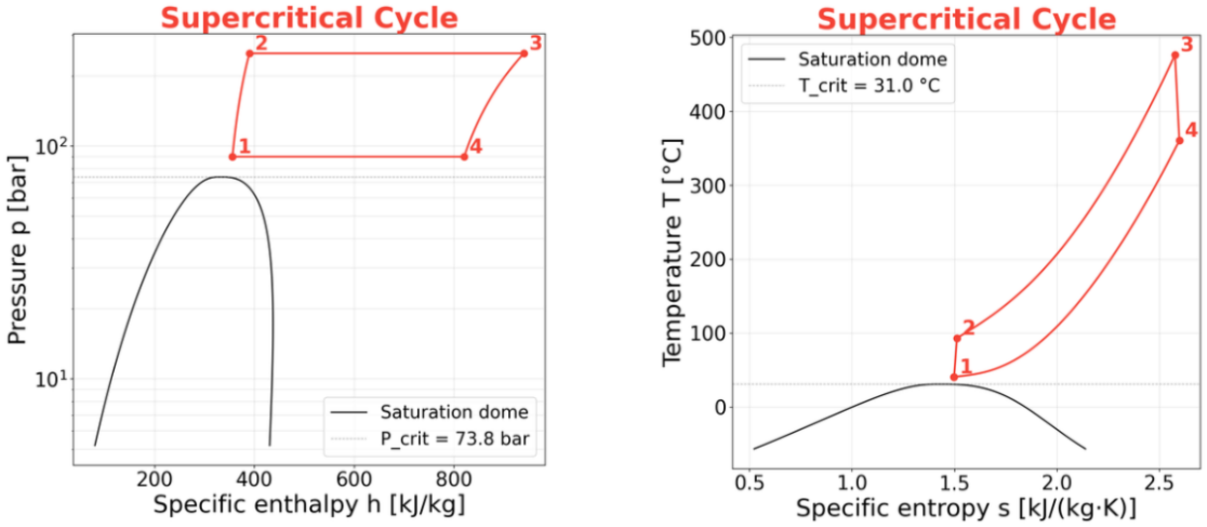


Figure 5: Diagram T-s and P-h of a Supercritical Cycle

Transcritical Cycles
A transcritical cycle occupies an interesting middle position. In this configuration, the fluid operates above its critical point on the high-pressure side (without phase change where the fluid behaves as a dense supercritical fluid) and below its critical point on the low-pressure side (where conventional evaporation and condensation occur).

An advantage of transcritical CO_2 cycles is the reduction of exergy destruction, leading to higher cycle efficiency. Exergy represents the maximum useful work obtainable as a system reaches equilibrium with its surroundings and exergy destruction is proportional to entropy generation. In heat exchangers, this entropy generation is largely caused by temperature differences between the heat source and the working fluid. In subcritical cycles, heat is added at constant temperature during phase change, which creates a mismatch when paired with sensible heat sources that cool continuously. This mismatch leads to large temperature differences, high irreversibility, and significant loss of work potential [18].

In contrast, transcritical CO_2 cycles operate above the critical pressure, where the fluid heats up continuously with a temperature glide. Near the critical point, CO_2 exhibits strong variations in thermophysical properties, especially in specific heat capacity, allowing its temperature profile to better match that of the heat source as shown in Figure 6. This thermodynamic matching maintains a smaller and more uniform temperature difference throughout the heat exchanger, reducing entropy generation and exergy destruction. As a result, more useful energy is preserved within the cycle, improving the overall efficiency of systems such as Carnot batteries.

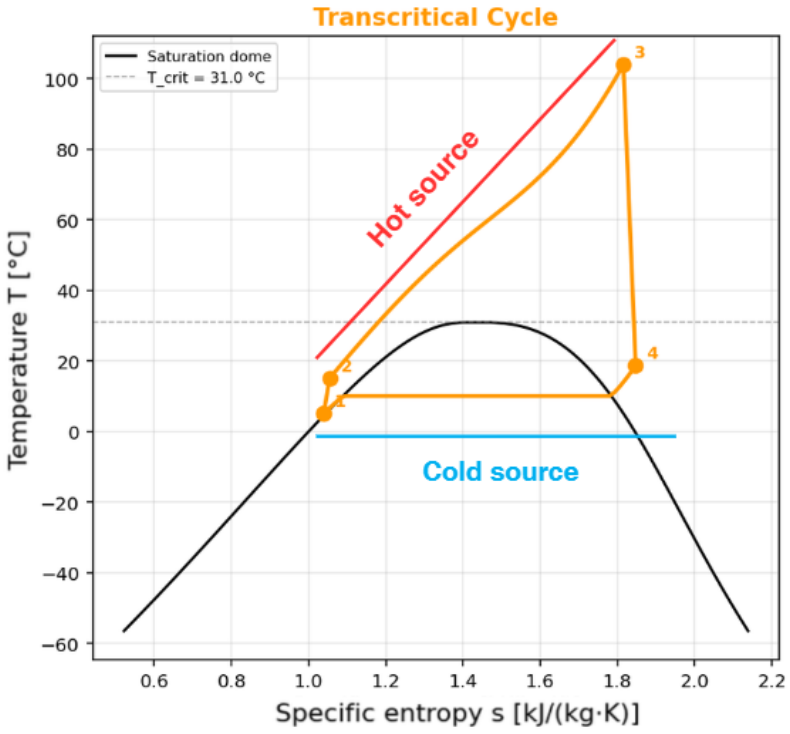


Figure 6: Temperature-entropy (T-s) diagram of the transcritical CO_2 cycle highlighting the thermodynamic matching.

Therefore, the result is a cycle that can achieve higher efficiency than a subcritical cycle at moderate temperatures, without the need for the extreme conditions of a purely supercritical cycle.

For CO_2 specifically, the transcritical configuration is attractive because its critical point (31.1 °C, 7.38 MPa) is low enough so that the low pressure side can condense at ambient/near-ambient temperatures using commonly available heat sinks (cooling water, air), while the high-pressure side operates at pressures of 80-160 bar and temperatures typically ranging from 70

to 200 ^{o}C for low-grade waste-heat applications, well within the range of standard industrial components.

2.2.3 Cycle Variants and Configurations

Within the broad category of transcritical CO_2 cycles, a number of architectural variants have been proposed and studied in the literature, each with a different trade-off between complexity, efficiency and cost:

Basic cycle

The simplest configuration consists of four components: a pump which pressurizes the liquid CO_2, a gas heater or evaporator which heats the CO_2 on the high-pressure side, an expander which extracts work as the fluid drops in pressure and a condenser which cools and condenses the fluid back to liquid on the low-pressure side. This is the baseline configuration, broadly equivalent to a simple Rankine cycle.

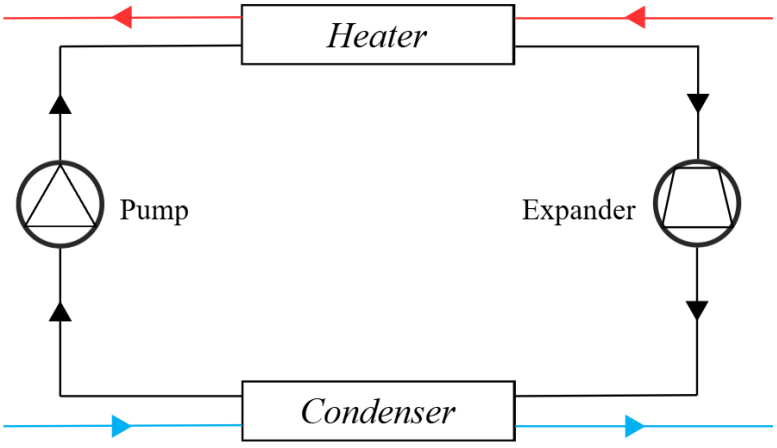


Figure 7: Basic Cycle Scheme

Recuperator cycle

A recuperator transfers heat from the hot expander outlet to the cooler pump outlet, preheating the high-pressure fluid before entering the gas heater. This reduces the heat input required and can increase the cycle efficiency, though its benefit depends strongly on operating conditions: in particular, when the expander outlet temperature is close to the high-pressure fluid inlet temperature, the recoverable heat becomes negligible and the pressure drops introduced by the recuperator can actually penalise the net efficiency [19]. The recuperator architecture was selected for this project because it offers the most cost-effective approach to improve cycle efficiency. By recovering sensible heat from the expander outlet to preheat the high-pressure fluid before it enters the heater, the required thermal input from the storage is reduced. This configuration provides better performance without introducing additional control complexities, as confirmed for transcritical CO_2 cycles in comparable low-temperature applications [20, 21].

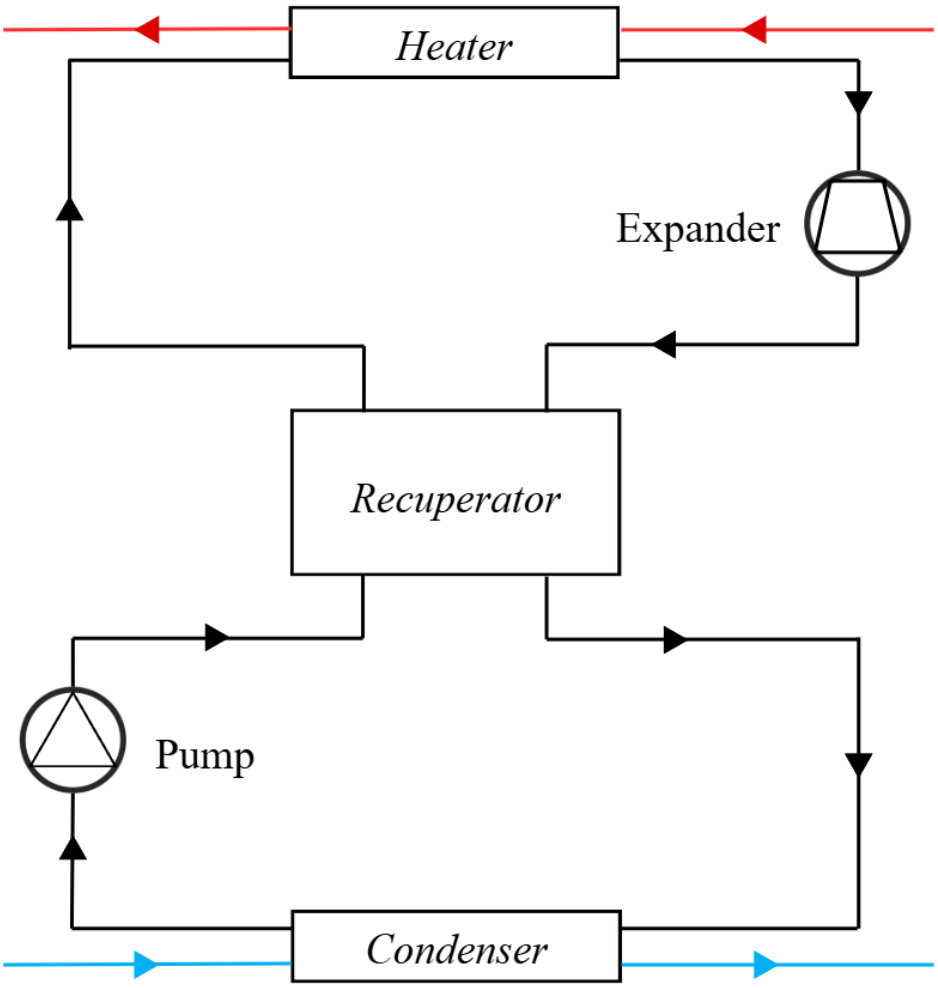


Figure 8: Recuperator Cycle Scheme

Recompression cycle

By adding a second compression stage or a recompressor that bypasses part of the flow, the system can better balance the thermal capacities of the hot and cold streams inside the recuperator, eliminating the internal pinch point that otherwise limits heat recovery. Specifically, this architecture was extensively developed for supercritical CO_2 (sCO_2) Brayton cycles to address a fundamental thermodynamic challenge: the variation in the specific heat capacity (c_p) of CO_2 near its critical point. In a simple recuperated cycle, the high-pressure cold stream has a much higher specific heat than the low-pressure hot stream, creating a thermal "pinch point" that severely limits heat recovery [16, 3].

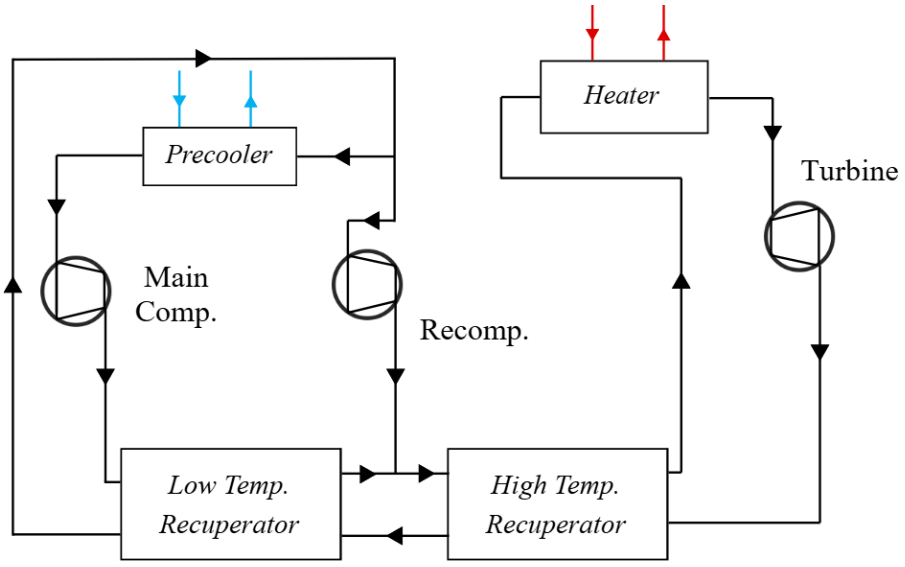


Figure 9: Recompression Cycle Scheme

The recompression layout solves this by splitting the recuperation process into a High-Temperature (HTR) and a Low-Temperature Recuperator (LTR). After the LTR, the flow is split: the main fraction travels through the cooler and the main compressor (operating near the critical point, where fluid density is highest, to minimise compression work). The remaining fraction bypasses the cooler and is routed directly to a secondary compressor (the recompressor). By reducing the mass flow in the cold side of the LTR, thermal capacities are balanced, thus eliminating the pinch point. The two streams are then mixed before entering the HTR. While these configurations can achieve significantly higher thermal efficiencies, they do so at the cost of increased system complexity, requiring additional turbomachinery and challenging control dynamics for the split fraction.

2.2.4 Challenges

Despite their advantages, transcritical CO_2 cycles also present challenges that must be understood before any control strategy can be designed:

High operating pressures: The transcritical cycle operates at high-side pressures of 8-20 MPa (80-200 bar), which is from 5 to 10 times higher than the pressures of conventional ORC systems. This imposes strict requirements on component design: thicker heat exchangers, more robust seals, heavier piping and fittings, among other implications. It also means that pressure control is critical for safety because an uncontrolled pressure increase could damage critical components.

Nonlinear properties near the critical point: As discussed in Section 2.2.1, the thermodynamic properties of CO_2 vary sharply near the critical point. This sensitivity means that the operating point and cycle efficiency respond strongly to small variations in temperature and pressure and a small disturbance in the heat source can produce disproportionate effects on system behaviour. Standard linear control techniques, which assume approximately constant system dynamics, may therefore perform poorly or become unstable in this regime [8].

Charge management: Previous studies show that in transcritical CO_2 cycles, system performance is strongly linked to refrigerant charge because it directly influences the optimal high-side pressure and thus overall efficiency [22]. Experimental and modelling work demonstrate that there is a clear optimum charge at which maximum efficiency is achieved, and deviations from this point (undercharging or overcharging) degrade performance.Undercharged operation reduces liquid inventory in key components such as flash tanks, limiting heat transfer and system capacity, while overcharging shifts operating pressure away from its optimum, also reducing efficiency.

2.3 Control of CO_2 Power Cycles

A Carnot Battery consists of a charging stage (Heat Pump) and a discharging stage (Power Cycle). This thesis focuses exclusively on the Power Cycle for a main reason. While high-temperature heat pumps are already well-documented for industrial decarbonization, the discharging stage (specifically using transcritical CO_2) remains less explored in current literature.

To understand the specific challenges of controlling a transcritical CO_2 power cycle, it is useful to first examine the control of a simpler baseline system, such as a conventional subcritical ORC, and then extend the analysis to its more advanced counterpart, namely supercritical cycles. This progressive approach allows for the accurate identification of the transcritical cycle's control variables, as the transcritical cycle can be considered an intermediate regime between subcritical and supercritical systems. These variables will be studied in detail in Chapter 3 and formally defined for control purposes in Chapter 4.

2.3.1 Control of Subcritical ORC Systems

In a subcritical ORC, the main manipulated variable is typically the pump speed N_{pump}. By increasing the pump speed, more fluid is sent to the evaporator, which raises the evaporation pressure and temperature. The key controlled variables are:

- **Evaporation temperature (T_{ev}):** Evaporation temperature is chosen to maximize the cycle efficiency given the heat source temperature. Running the evaporator too hot wastes potential work; running it too cold risks incomplete evaporation.

- **Superheat at the expander inlet (ΔT_{sh}):** This is a safety parameter. The fluid entering the expander must be fully in the vapour phase; any liquid droplets would damage the turbine blades. A minimum positive superheat (typically between 5 and 10 K above the saturation temperature) must be maintained. However, excessive superheat reduces efficiency because it represents energy that does not contribute to increasing the high-pressure level and therefore does not translate into additional turbine work, beyond what is needed for safe operation.
- **Hot-fluid mass flow rate ($\dot{m}_{hf}$):** The mass flow rate of the hot fluid entering the gas heater acts as a boundary condition that determines the maximum available heat input and thus the optimal evaporation temperature. In configurations where this flow can be adjusted (e.g. via a pump on the heat transfer fluid loop), it also becomes an additional manipulated variable for modulating the thermal power delivered to the cycle.

The relatively stable thermodynamic behaviour of subcritical fluids means that PID (proportional-integral-derivative) controllers can often achieve satisfactory performance in subcritical ORC systems, at least near the design point.[23]

2.3.2 Control of Supercritical CO_2 Brayton Cycles

Research has also been carried out on the control of supercritical CO_2 Brayton cycles (sCO_2) for power generation applications [3]. In sCO_2 systems, the main manipulated variables are typically the turbomachinery speeds (N_{comp}, N_{exp}). Varying the rotational speed of the turbine and compressor provides a primary degree of freedom to adjust mass flow rates, pressure ratios, and the overall power output of the cycle. Furthermore, as highlighted in the review [3], other critical manipulated variables include inventory control (M_{tot}) by injecting or extracting CO_2 mass from the main loop via storage tanks to modify operating pressures and fluid density at part-load, and the use of bypass valves (such as turbine or recuperator bypasses, which allow for rapid power adjustments and temperature regulation during transients).

By acting on these manipulated variables, the system can regulate the key controlled variables, which provide relevant information for the transcritical case due to the similarities at the high-pressure side. The key controlled variables are:

- **Main Compressor Inlet Pressure/Temperature (MCIP/MCIT):** Operating the compressor inlet as close as possible to the critical point of CO_2 is thermodynamically advantageous because the fluid density is maximised there, minimising compression work. However, if the temperature falls too low, the system risks entering the subcritical (two-phase) region, leading to condensation inside the compressor which is a condition that can cause mechanical damage. Robust control of the compressor inlet conditions is therefore essential both for performance and safety.

- **Turbine Inlet Temperature (TIT):** The maximum temperature of the working fluid entering the turbine determines the cycle's thermodynamic efficiency. A higher TIT yields a larger enthalpy drop across the turbine and consequently more work output. In practice, however, the TIT is constrained by material limits. While advanced cooling techniques allow gas turbine blades to operate at metal temperatures exceeding 1000 °C in open-air cycles, CO_2 introduces an additional constraint: it becomes increasingly corrosive to structural alloys (Fe- and Ni-based) above approximately 450–650 °C, through mechanisms such as carburisation, oxidation and creep [24]. The TIT therefore represents a fundamental trade-off between thermodynamic performance and equipment longevity.

2.4 Control Challenges Specific to the Transcritical CO_2 Rankine Cycle: Key Control Parameter Identification

Based on the subcritical ORC and sCO_2 Brayton literature, and integrating the specific characteristics of the transcritical cycle, the key variables of the transcritical CO_2 Rankine power block are classified into two categories: manipulated variables, which are the degrees of freedom available to the control system, and controlled variables, which are the quantities whose values must be regulated to ensure safe and efficient operation. The sensitivity analysis in Section 3 will verify and quantify their relative importance.

Manipulated Variables

1. **Hot-source mass flow rate ($\dot{m}_{hf}$).** The mass flow rate of the heat transfer fluid governs the thermal power delivered to the gas heater:

$$\dot{Q}_{in} = \dot{m}_{hf} \cdot c_p \cdot (T_{in} - T_{out}) \tag{2}$$

   Varying $\dot{m}_{hf}$ therefore modulates the energy input to the cycle, directly affecting both the turbine inlet conditions and the net power output:

$$\dot{m}_{hf} \uparrow \Rightarrow \ \dot{Q}_{in} \uparrow \Rightarrow \ \dot{W}_{net} \uparrow \tag{3}$$

2. **Pump rotational speed (N_{pp}).** The pump speed is the primary fast actuator for regulating the CO_2 mass flow rate $\dot{m}$ through the cycle. An increase in pump speed raises the mass flow rate, which directly increases the thermal power absorbed in the gas heater and the net power output:

$$\dot{Q}_{gh} = \dot{m} \cdot (h_{out} - h_{in}) \tag{4}$$

$$N_{pp} \uparrow \Rightarrow \ \dot{m} \uparrow \Rightarrow \ \dot{W}_{net} \uparrow \tag{5}$$

If P_{high} is maintained approximately constant via inventory control, the specific work w_{net} remains nearly unchanged, and the power output scales proportionally with mass flow rate [23].

$$\dot{W}_{net} = \dot{m} \cdot w_{net} \tag{6}$$

3. **Expander rotational speed (N_{exp}).** In configurations where the expander speed is decoupled from the grid frequency (e.g., via a variable-speed generator), N_{exp} provides an additional degree of freedom. Adjusting the expander speed modifies the pressure ratio across the turbine and consequently, the net power output and the high-side pressure:

$$N_{exp} \uparrow \Rightarrow \ \dot{W}_{net} \uparrow \tag{7}$$

4. **Cooling-fluid mass flow rate ($\dot{m}_{cf}$).** On the low-pressure side of the cycle, the mass flow rate of the cooling fluid circulating through the condenser determines the heat rejection capacity. A higher cooling flow rate lowers the condensation temperature and, consequently, reduces the low-side pressure P_{low}:

$$\dot{m}_{cf} \uparrow \Rightarrow \ T_{cond} \downarrow \Rightarrow \ P_{low} \downarrow \tag{8}$$

This makes $\dot{m}_{cf}$ the natural actuator for regulating P_{low}, which in turn influences the overall pressure ratio and cycle efficiency.

Controlled Variables

1. **High-side pressure (P_{high}).** The single most influential variable,as it couples thermodynamic performance to fluid inventory. Both too low and too high values reduce efficiency: too low pressure means insufficient enthalpy drop across the expander; too high pressure increases pump work and moves the cycle away from the optimal thermodynamic region near the critical point [8, 22].

2. **Turbine inlet temperature (TIT).** The TIT defines the maximum temperature level of the cycle and sets the upper limit of the energy convertible into useful work. In a transcritical CO_2 Rankine cycle, the TIT is determined by the hot-source temperature and the pinch-point constraint in the gas heater:

$$TIT \approx T_{hf} - \Delta T_{min} \tag{9}$$

A higher TIT yields a larger enthalpy drop across the expander and therefore greater specific work output:

$$TIT \uparrow \Rightarrow\ h_{TIT} \uparrow \Rightarrow\ w_{net} \uparrow \Rightarrow\ \dot{W}_{net} \uparrow \tag{10}$$

Studies have shown that, within typical operating ranges (70 to 200 °C), increasing TIT leads to a consistent increase in thermal efficiency [8]. The TIT is therefore monitored as a key performance indicator and safety boundary.

3. **Net power output ($\dot{W}_{net}$).** The net power output is the primary performance objective of the power block. Regulating $\dot{W}_{net}$ to match a grid demand signal or dispatch setpoint is the top-level control objective. It is influenced by all other manipulated variables, making it a suitable output variable for supervisory control.

4. **Low-side pressure (P_{low}).** The condensation pressure on the low-pressure side of the cycle is determined by the heat rejection conditions at the condenser. It directly sets the back-pressure against which the expander exhausts and therefore influences the overall pressure ratio and the net work output:

$$P_{low} \downarrow \Rightarrow\ \frac{P_{high}}{P_{low}} \uparrow \Rightarrow\ w_{net} \uparrow \tag{11}$$

Controlling P_{low} also ensures that the condensation temperature stays sufficiently above the cooling-fluid inlet temperature to maintain the required pinch-point margin, which could cause operational instabilities in the condenser. P_{low} is therefore a secondary but important controlled variable, regulated via $\dot{m}_{cf}$.

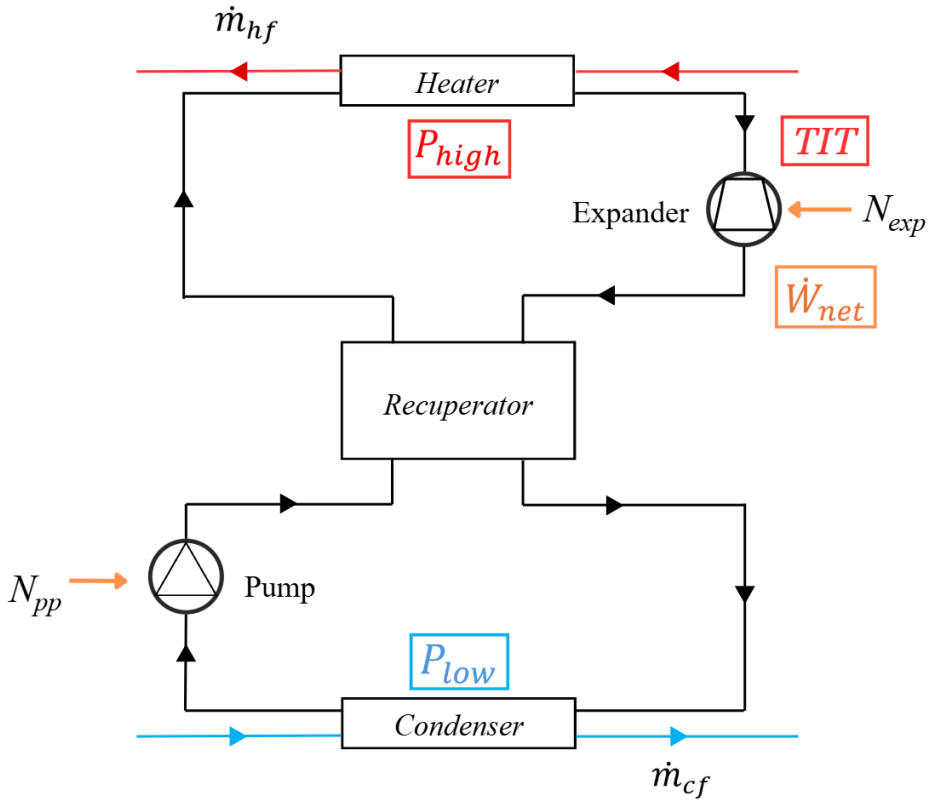


Figure 10: Schematic of the transcritical CO_2 Rankine cycle highlighting the manipulated variables (black) and controlled variables (colored)

The literature reviewed in this chapter shows that transcritical CO_2 power cycles have been investigated from complementary perspectives. Thermodynamic optimisation has been addressed for low-grade waste-heat recovery [25, 18, 2] and for low-temperature geothermal applications [15, 20], while component-level studies have quantified the influence of recuperator design and effectiveness on overall cycle performance [21, 20]. More recently, the adoption of transcritical CO_2 cycles within Carnot battery and pumped thermal energy storage configurations has been explored at MW-class scales [26, 27, 6]. In parallel, control-oriented analyses have been extensively developed for subcritical ORC systems [23, 14] and for supercritical CO_2 Brayton cycles [3, 16].

However, three gaps remain. First, most research on transcritical CO_2 cycles focuses on heat pump applications, while the power cycle configuration has received less attention, despite its relevance in Carnot battery systems where both charge and discharge stages affect round-trip efficiency. Second, although control strategies for supercritical CO_2 Brayton cycles are well documented [3], the transcritical Rankine cycle operates in a different thermodynamic regime and has been less studied from a control perspective. Third, there is a lack of performance

maps linking boundary conditions to optimal operating points (P_{high}, $\dot{m}$, $\dot{m}_{hf}$).

To address these gaps, the present thesis performs a control-oriented thermodynamic analysis of a recuperated transcritical CO_2 Rankine cycle for Carnot battery applications. As a first step, the manipulated and controlled variables identified in this chapter are quantified through an on-design sensitivity analysis (Chapter 3) which provides performance maps over a relevant range of source temperatures. These maps confirm or refine the proposed control variables and define the reference setpoints used in the part-load study and control architecture developed in Chapter 4.

3 Thermodynamic modelling and On-Design Study

3.1 Modelling assumptions

The goal of this chapter is to characterise the thermodynamic behaviour of the recuperated transcritical CO_2 Rankine power block under on-design conditions. The analysis produces performance maps: for each combination of hot-source temperature (T_{hf}) and cold-source temperature (T_{cf}), the model determines the highest achievable net cycle efficiency and the corresponding optimal high-side pressure P_{high}, CO_2 mass flow rate $\dot{m}$ and hot-source mass flow rate $\dot{m}_{hf}$.

The model is implemented in Python using the *LaboThApPy* library developed at the University of Liège (ULiège). LaboThApPy provides a modular, component-based framework in which each piece of equipment (pump, gas heater, recuperator, expander, condenser) is modelled as an independent thermodynamic block connected through mass connectors. Fluid property calculations are handled by the *CoolProp* library, which provides the equations of state for CO_2 across the operating range.

The following modelling decisions define the scope of this study:

- **Steady-state only:** No dynamic effects are considered. This is a standard approach for on-design studies and is sufficient to generate the performance maps used to inform the control-architecture analysis of Chapter 4.
- **No internal component geometry:** The model does not require knowledge of heat-exchanger surface areas, turbomachinery blade profiles or pipe diameters. Component performance is characterised through efficiency parameters (isentropic efficiency, heat-exchanger effectiveness, pinch-point temperature difference). This makes the model both fast to evaluate and applicable even before the physical hardware is specified.
- **Recuperated cycle architecture:** The configuration is the recuperated layout described in Section 2.2.3, selected as the best balance between performance and control complexity.
- **Discretized heat exchanger modelling:** Due to the transcritical nature of CO_2, operating pressures are close to the critical point (73.8 bar, 31.1 °C). In this region, thermophysical properties, particularly the specific heat capacity (c_p), exhibit highly nonlinear spikes with temperature (Section 2.2.1). To capture these variations accurately and prevent physically impossible solutions (such as internal temperature crossovers), the gas heater and the recuperator are divided into n_{disc} discrete volumes along their thermal length.

- **Boundary temperature ranges:** The sweep covers $T_{hf} \in [130, 200]\,^{o}\mathrm{C}$ on the hot side and $T_{cf} \in [0.1, 20]\,^{o}\mathrm{C}$ on the cold side. The hot-source range sits inside the upper band of the low-grade waste-heat category ($< 240\,^{o}\mathrm{C}$) examined by Wolf et al. [2] in their comparative study of ten transcritical CO_2 architectures, and is the regime where this kind of cycle is known to outperform conventional ORCs thanks to temperature-glide matching [18]. The cold-source range spans from ambient cooling-water conditions ($\approx$ 15–20 $^{o}\mathrm{C}$) down to the near-freezing temperatures attainable with refrigerated cooling loops or with dedicated low-temperature thermal storage units. Ice-based cold reservoirs are explicitly proposed for transcritical CO_2 Carnot batteries by Steinmann et al. [6], who couple a transcritical CO_2 Rankine cycle with an ice storage unit on the cold side, allowing condensation very close to $0\,^{o}\mathrm{C}$.

- **Fixed net power target:** A net electrical output of $\dot{W}_{net} = 10\,\mathrm{MW}$ is imposed throughout the analysis. This figure reflects the scale of commercially relevant Carnot-battery discharge units currently being developed in the long-duration energy storage (LDES) sector. For example, Echogen is targeting prototype installations in the 10MWe range [10]. Adopting a fixed power target throughout the analysis ensures a consistent basis for comparison between operating conditions. By enforcing the same net power output across the entire hot- and cold-source temperature domain, efficiency variations can be attributed solely to the influence of boundary conditions rather than differences in operating point.

- **Net cycle efficiency convention:** Throughout this chapter, the reported efficiency is defined as $\eta = \dot{W}_{net}/\dot{Q}_{gh}$, where $\dot{W}_{net}$ includes the auxiliary power consumed by the hot-source circulation pump (see Section 3.2). This is therefore a net-cycle metric rather than a strict thermal efficiency.

- **Cold source treated as effectively unlimited:** The cooling-water stream is set at a large mass flow rate ($\dot{m}_{cf} = 5000\,\mathrm{kg/s}$) so that its temperature rise across the condenser is negligible and T_{cf} behaves as a true boundary condition. The pumping power of the cold-source loop is not included in the net-power balance.

- **Saturated liquid at pump inlet:** A liquid CO_2 reservoir is assumed downstream of the condenser; the pump always sees saturated liquid conditions. A minimum subcooling of 1 K is imposed to prevent cavitation.

- **Stochastic optimisation:** A Particle Swarm Optimisation (PSO) algorithm finds the best thermodynamic operating point for each pair (T_{hf}, T_{cf}) (see Section 3.3 for details).

The general structure of the model is summarised in Figure 11. The inputs are the boundary temperatures (T_{hf}, T_{cf}); the fixed inputs are the component efficiencies, the pinch-point constraints and the 10 MW power target; the outputs are the net cycle efficiency η and the optimal operating point, defined by the high-side pressure P_{high}, the CO_2 mass flow rate $\dot{m}$ and the hot-source mass flow rate $\dot{m}_{hf}$.

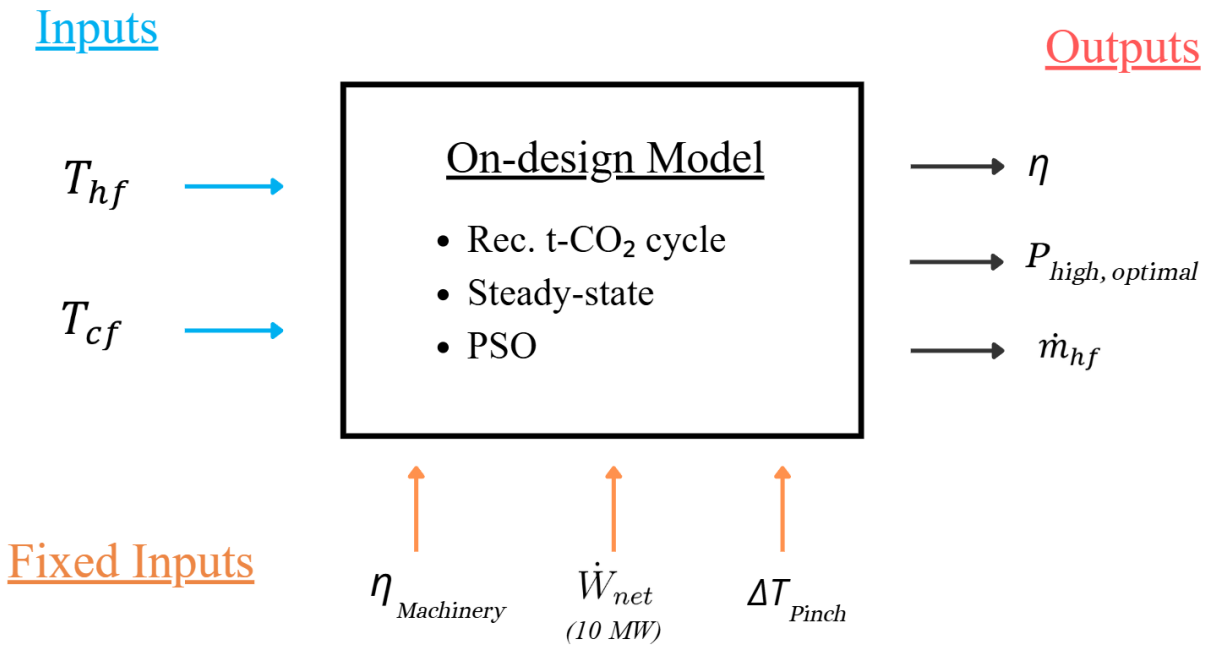


Figure 11: On-design model structure: boundary temperatures as inputs, fixed component parameters, and optimal operating point as outputs.

3.2 Component models and parameters

Each component of the recuperated transcritical CO_2 cycle is modelled under the assumptions detailed below. The fixed performance parameters are summarised in Table 1 and have been chosen in line with values consistently adopted in the transcritical CO_2 Rankine literature for low-grade waste-heat recovery and Carnot-battery applications. The corresponding sources are introduced component by component.

Pump. The CO_2 pump raises the fluid from the low-side condensation pressure P_{low} to the high-side pressure P_{high}. The actual pump work per unit mass is

$$w_{pp} = \frac{h_{out,s} - h_{in}}{\eta_{pp}}, \tag{12}$$

where $h_{out,s}$ is the isentropic outlet enthalpy and $\eta_{pp} = 0.85$ is the isentropic efficiency. The same value is adopted by Chen et al. [25] in their foundational comparative study of the transcritical CO_2 power cycle against an R123-based ORC for waste-heat recovery, and by Yu et al. [27] in their recent feasibility analysis of a transcritical pumped thermal energy storage system bottoming at ambient temperature.

Gas heater. The gas heater transfers thermal energy from the hot-source fluid (pressurised water) to the supercritical CO_2. A pinch-point temperature difference of $\Delta T_{gh} = 5\,\mathrm{K}$ is imposed as a minimum approach constraint, which sets the maximum achievable turbine inlet temperature:

$$TIT \leq T_{hf,in} - \Delta T_{gh}. \tag{13}$$

A heat-exchanger effectiveness of $\varepsilon_{gh} = 0.95$ is applied, and pressure drops on both sides are accounted for ($\Delta P_h = 0.5\,\mathrm{bar}$, $\Delta P_c = 2\,\mathrm{bar}$). This effectiveness matches the value used by Baheta et al. [21] in their parametric first- and second-law analysis of regenerative transcritical CO_2 Rankine cycles for low-grade heat conversion. The 5 K minimum pinch follows the assumption adopted by Yu et al. [27] for the heat exchangers of their large-scale transcritical PTES system.

Recuperator. The recuperator recovers sensible heat from the expander outlet to preheat the high-pressure CO_2 before the gas heater, reducing the required heat input. An effectiveness of $\varepsilon_{rec} = 0.80$ is used, with pressure drops of $\Delta P_h = 1\,\mathrm{bar}$ and $\Delta P_c = 2\,\mathrm{bar}$.As in the gas heater, a discretised effectiveness model is applied to detect any internal temperature cross arising from the non-linear c_p profile of the high-pressure CO_2 stream, and the effectiveness is iteratively reduced from $\varepsilon_{rec} = 0.80$ until the minimum internal pinch is satisfied at every segment. The chosen value sits in the central band of effectiveness levels investigated by Wu et al. [20], who specifically studied the influence of recuperator parameters on the design and off-design performance of transcritical CO_2 Rankine cycles for low-temperature geothermal applications.

Expander. The expander extracts work as the supercritical CO_2 expands from P_{high} to P_{low}. An isentropic efficiency of $\eta_{exp} = 0.90$ is applied:

$$w_{exp} = \eta_{exp} \cdot (h_{in} - h_{out,s}), \tag{14}$$

in line with the turbine efficiency used by Chen et al. [25] for the transcritical CO_2 cycle. A liquid-fraction check is performed at the expander outlet: if condensation is detected (outlet superheat $\Delta T_{sh} < 0$ K), the evaluation is penalised and rejected, since liquid droplets in the expander would cause mechanical damage.

Condenser. The condenser rejects heat to the cold source. A minimum pinch-point temperature difference of $\Delta T_{cd} = 5\,\mathrm{K}$ is imposed between the cold-source inlet and the condensation temperature, matching the $5\,^{\mathrm{o}}\mathrm{C}$ pinch adopted by Yu et al. [27] for the heat exchangers of their large-scale transcritical PTES analysis. A subcooling of 1 K is imposed at the outlet to ensure fully liquid conditions at the pump inlet. Pressure drops of $\Delta P_h = 1\,\mathrm{bar}$ and $\Delta P_c = 0.5\,\mathrm{bar}$ are included.

Auxiliary pump work. The pumping power required to circulate the hot-source fluid through the gas heater is subtracted from the expander output to avoid overestimating system efficiency:

$$\dot{W}_{aux} = \frac{\Delta P_{h,gh} \cdot \dot{m}_{hf}}{\rho_{hf} \cdot \eta_{pp}}. \tag{15}$$

The net power output is therefore

$$\dot{W}_{net} = \dot{W}_{exp} - \dot{W}_{pp} - \dot{W}_{aux}. \tag{16}$$

The cold-source pumping power is not included in the balance, in line with the assumption of an effectively unlimited cooling supply stated in Section 3.1.

Table 1: Fixed model parameters used in the on-design study.

Component	Parameter	Symbol	Value
Pump	Isentropic efficiency	η_{pp}	0.85
Gas heater	Effectiveness	ε_{gh}	0.95
	Min. pinch point	ΔT_{gh}	5 K
Recuperator	Effectiveness	ε_{rec}	0.80
Expander	Isentropic efficiency	η_{exp}	0.90
Condenser	Min. pinch point	ΔT_{cd}	5 K
	Subcooling	SC_{cd}	1 K
System	Target net power	$\dot{W}_{obj}$	10 MW

3.3 On-design optimisation methodology

For each pair of boundary temperatures (T_{hf}, T_{cf}), the on-design problem consists in finding the operating point that maximises the net cycle efficiency while delivering the prescribed power output. The decision vector is

$$\mathbf{x} = (P_{high},\ \dot{m},\ \dot{m}_{hf}), \tag{17}$$

and the optimisation problem reads

$$\max\ \eta(\mathbf{x}) \quad \text{s.t.} \quad \dot{W}_{net}(\mathbf{x}) = 10\,\text{MW}, \tag{18}$$

with bounds set from preliminary scoping runs: $P_{high} \in [130, 250]$ bar, $\dot{m} \in [170, 470]$ kg/s, and the factor $\dot{m}_{hf}/\dot{m} \in [0.1, 2.0]$. The power constraint is enforced through a penalty when the deviation exceeds 1 %; thermodynamically unstable candidates (negative superheat at the expander outlet, recuperator pinch violation, solver divergence) are penalised so the optimiser discards them automatically.

The objective function $\eta(\mathbf{x})$ is non-convex and non-differentiable: CO_2 properties vary strongly and non-linearly near the critical point, the cycle solver is iterative, and the penalty function introduces discontinuities at infeasible boundaries. Gradient-based methods are therefore unsuitable, as they require a smooth, differentiable landscape and are prone to becoming trapped in local optima when these conditions are not met. A *Particle Swarm Optimisation* (PSO) metaheuristic is adopted instead. PSO explores the search space with a population of candidate solutions that exchange information and converge iteratively towards the best region found, without ever computing derivatives [28]. This makes it robust to the discontinuities and non-linearities inherent in the present problem. The implementation uses the open-source *pyswarms* library [29].

The performance maps are obtained by sweeping the PSO over a two-dimensional grid in (T_{cf}, T_{hf}). To mitigate the stochastic nature of the algorithm, each optimisation is repeated five times with independent random initialisations and the solution with the highest net efficiency is retained. The result is, for every (T_{cf}, T_{hf}) pair, a converged optimum $(P_{high}, \dot{m}, \dot{m}_{hf})$ delivering $\dot{W}_{net} = 10\,\mathrm{MW}$ within 1 % tolerance. Collecting these optima across the grid produces the four performance maps analysed in Section 3.4.

3.4 On-design results and performance maps

The on-design optimisation is run on a grid covering $T_{cf} \in [0.1, 20]\,^\circ\mathrm{C}$ and $T_{hf} \in [130, 200]\,^\circ\mathrm{C}$. The global optimum across the explored space is located at the Carnot-favoured corner ($T_{hf} = 200\,^\circ\mathrm{C}$, $T_{cf} = 0.1\,^\circ\mathrm{C}$), while delivering the prescribed 10 MW of net power. The corresponding operating conditions are summarised in Table 2.

Table 2: Global on-design optimum identified by the PSO across the (T_{cf}, T_{hf}) search space.

Quantity	**Symbol**	**Value**
Hot-source temperature	T_{hf}	$200\,^\circ\mathrm{C}$
Cold-source temperature	T_{cf}	$0.1\,^\circ\mathrm{C}$
Net cycle efficiency	η	20.3 %
High-side pressure	P_{high}	218 bar
CO_2 mass flow rate	$\dot{m}$	161 kg/s
Hot-source mass flow rate	$\dot{m}_{hf}$	80 kg/s
Net power output	$\dot{W}_{net}$	10 MW

The four performance maps that follow are interpreted below, in the order: net cycle efficiency, CO_2 mass flow rate, hot-source mass flow rate and high-side pressure.

3.4.1 Net cycle efficiency map

The net cycle efficiency map (Figure 12) shows the variation of η as a function of the hot- and cold-source temperatures. The efficiency increases with T_{hf} and decreases with T_{cf}, reaching a maximum of 20.3 % at the high-T_{hf}, low-T_{cf} corner.

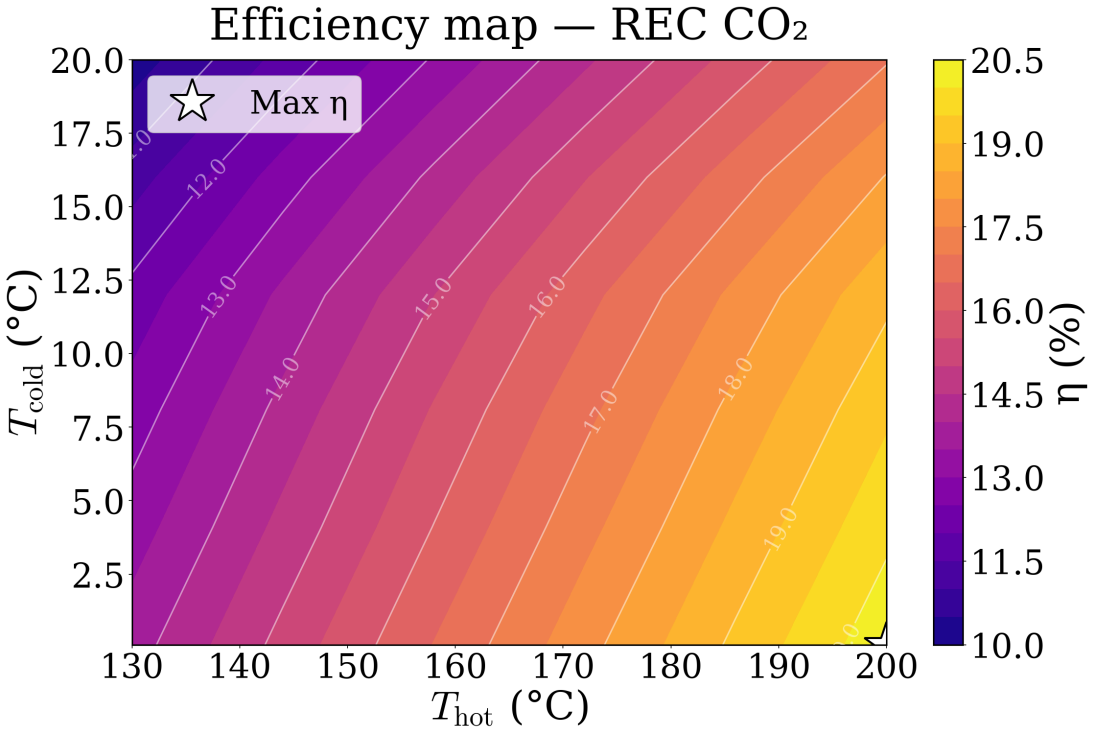


Figure 12: Net cycle efficiency η of the recuperated transcritical CO_2 cycle as a function of T_{hf} and T_{cf}, at a fixed net power of 10 MW. The white star marks the global optimum.

Carnot bound. The trend is consistent with the Carnot upper bound,

$$\eta_{Carnot} = 1 - \frac{T_{cf}}{T_{hf}}, \tag{19}$$

where temperatures are expressed in Kelvin and favours hotter sources and colder sinks. Across the explored range, η_{Carnot} varies between approximately 27 % (at $T_{hf} = 130\,°C$, $T_{cf} = 20\,°C$) and 42 % (at $T_{hf} = 200\,°C$, $T_{cf} = 0.1\,°C$). The actual cycle efficiency reaches 20.3 % at the optimum, i.e. roughly 48 % of the corresponding Carnot limit.The gap between this Carnot upper bound and the actual cycle efficiency is a direct consequence of the realistic component assumptions adopted in Table 1. Together, these irreversibilities account for the reduction from the Carnot limit to the simulated 20.3%.

Role of the cold sink. A lower T_{cf} reduces the condensation pressure P_{low}, which increases the pressure ratio across the expander and therefore the specific enthalpy drop. This explains the monotonic improvement of η towards lower T_{cf} values.

Role of the hot source. A higher T_{hf} raises the turbine inlet temperature TIT, which is bounded by the gas-heater pinch constraint, and consequently enlarges the enthalpy drop across the expander. The turbine inlet temperature was identified in Section 2.4 as a primary controlled variable, in line with its role here as a key driver of cycle performance.

3.4.2 Optimal CO_2 mass flow rate map

The CO_2 mass flow rate map (Figure 13) follows the same trend as the efficiency map: at the high-T_{hf}, low-T_{cf} corner the required $\dot{m}$ drops to 161 kg/s, while at the opposite corner it rises to about 465 kg/s, a factor of roughly three across the operating space.

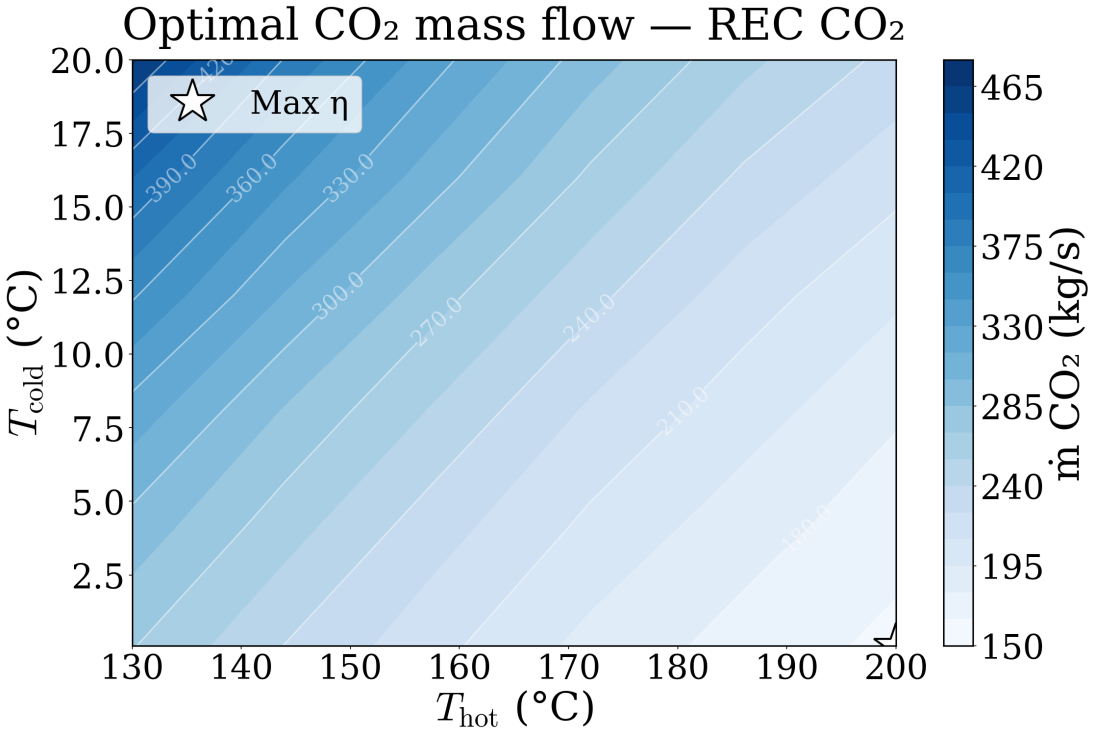


Figure 13: Optimal CO_2 mass flow rate $\dot{m}$ as a function of T_{hf} and T_{cf}.

Fixed-power constraint. A hotter source and a colder sink increase the temperature difference across which the cycle operates, so each kilogram of CO_2 carries more useful energy through the expander. Since the net power output is fixed at 10 MW, a higher specific work per

kilogram directly translates into a lower mass flow rate. More formally, the net specific work of the cycle is

$$w_{net} = \big(h_{TIT} - h_{exp,out}\big) - \big(h_{pp,out} - h_{pp,in}\big), \tag{20}$$

and the required mass flow rate to deliver $\dot{W}_{net} = 10\,\mathrm{MW}$ is

$$\dot{m} = \frac{\dot{W}_{net}}{w_{net}}. \tag{21}$$

A higher T_{hf} raises the enthalpy drop across the expander, while a lower T_{cf} reduces P_{low} and increases the expansion ratio. Both effects increase w_{net} and therefore reduce the mass flow required to maintain 10 MW.

Practical implications. As discussed in Section 2.2.1, one of the main advantages of CO_2 as a working fluid is the compactness of the resulting turbomachinery and heat exchangers, driven by the high fluid density near the critical point. The mass-flow map quantifies this advantage: at the most favourable corner the system circulates approximately a factor of three less CO_2 than at the least favourable one. This translates into smaller pump and expander displacement, smaller heat-exchanger headers and pipework, lower auxiliary parasitic power and a lower capital cost on the CO_2 side of the plant.

Control implications. From a control perspective, $\dot{m}$ is the natural output of the pump-speed control loop (N_{pp} in Section 2.4). The mass-flow map defines the modulation range that the pumping system must cover, corresponding to an approximate 3:1 turndown ratio. This sets a design requirement on both the pump and its variable-frequency drive, which must operate stably across the full envelope.

3.4.3 Optimal hot-source mass flow rate map

The hot-source mass flow rate map (Figure 14) shows $\dot{m}_{hf}$ ranging from about 80 kg/s at the high side temperatures to about 350 kg/s at the opposite corner, a factor of roughly 4.4.

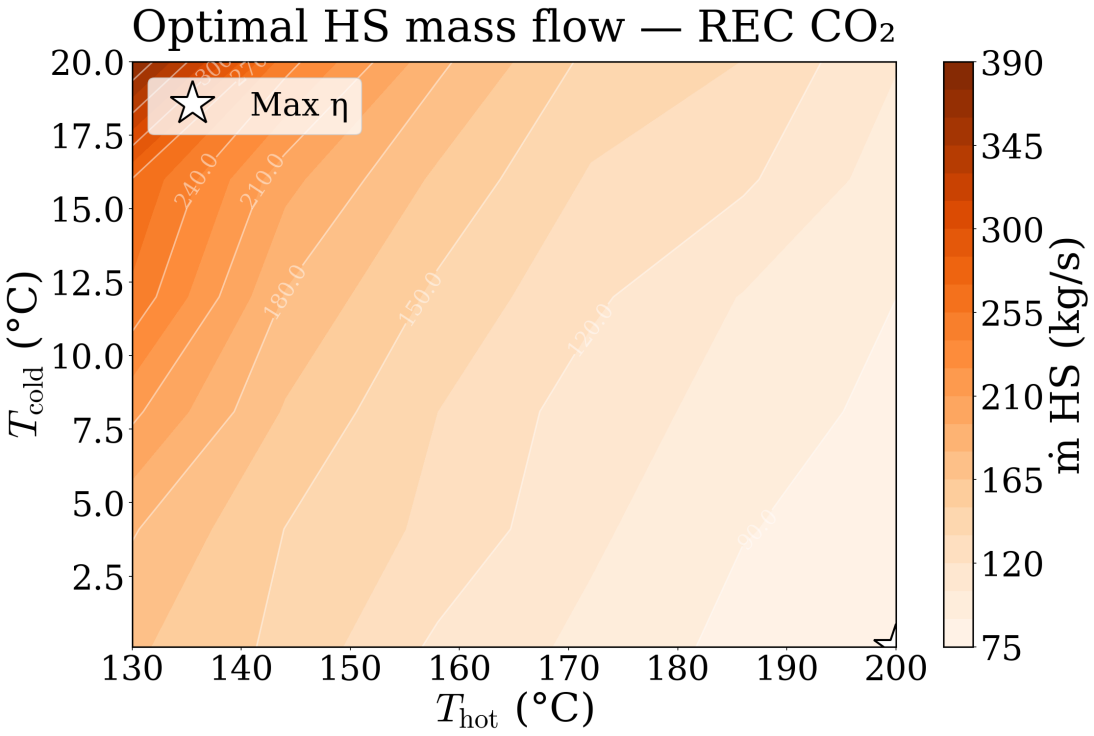


Figure 14: Optimal hot-source mass flow rate $\dot{m}_{hf}$ as a function of T_{hf} and T_{cf}.

Dependence on boundary temperatures. As T_{hf} increases, ΔT_{hf} grows because the hot fluid enters at a higher temperature while its outlet is bounded below by the gas-heater pinch constraint. A larger ΔT_{hf} means the same $\dot{Q}_{gh}$ can be delivered with less mass flow, and therefore $\dot{m}_{hf}$ drops with T_{hf}. The dependence on T_{cf} enters through η: a colder sink improves cycle efficiency, reduces the required $\dot{Q}_{gh}$, and therefore also reduces $\dot{m}_{hf}$. The combination of both effects produces the diagonal pattern observed in Figure 14, with T_{hf} as the dominant axis.

Practical implications. A lower $\dot{m}_{hf}$ reduces the required capacity of the hot-source pumps and piping infrastructure. It also decreases the auxiliary pumping power and the hydraulic losses on the hot-source side. The 4.4× variation across the map highlights the importance of $\dot{m}_{hf}$ as a manipulated variable: as identified in Section 2.4, modulating the hot-source flow in step with operating conditions is an effective way to regulate the heat input to the cycle while keeping the auxiliary consumption under control. This relationship will be exploited in the control architecture of Chapter 4.

3.4.4 Optimal high-side pressure map

The high-side pressure map (Figure 15) is the most complex of the four, because P_{high} results from several competing effects acting simultaneously. The general trend is a monotonic increase with T_{hf}, from approximately 145 bar at $T_{hf} = 130\,°C$ to 230 bar at $T_{hf} = 200\,°C$, with a secondary increase toward higher T_{cf}.

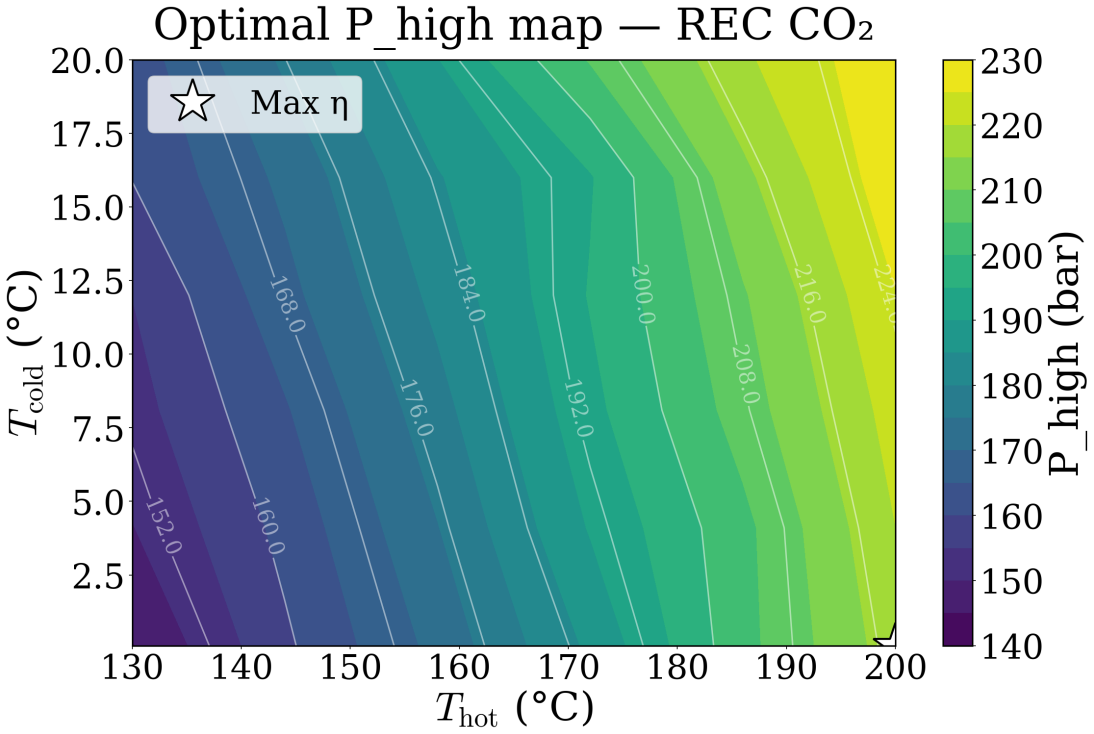


Figure 15: Optimal high-side pressure P_{high} as a function of T_{hf} and T_{cf}.

Trade-off between pump and expander. One of the reasons why an optimal pressure exists is the way P_{high} affects the two turbomachinery components. The pump work scales approximately linearly with the pressure difference,

$$\dot{W}_{pp} \approx \dot{m}\,\frac{P_{high} - P_{low}}{\rho_{liq}\,\eta_{pp}}, \tag{22}$$

whereas the specific work extracted by the expander grows more slowly with pressure ratio than the pump work, since the enthalpy drop across the expander increases sub-linearly as the expansion ratio rises in the operating regime considered.

Consequently, this results in a fixed increase in the work done by the pump, but generates an ever-decreasing increase in the work done by the expander. Beyond the optimal pressure

the pump penalty exceeds the expander gain, and net power decreases.

Temperature-glide matching. A higher T_{hf} requires a higher P_{high} to maintain effective heat transfer in the gas heater. As discussed in Section 2.2.1, near the critical point the specific heat of CO_2 varies strongly with temperature, distorting its heating profile. Operating at higher pressures moves the fluid away from the critical region, flattening the T-h curve and improving the match with the linearly cooling hot source. This matching reduces entropy generation and is the main driver of the strong dependence of P_{high} on T_{hf}.

Role of cold-source temperature. A warmer sink , i.e. higher T_{cf}, raises the condensation pressure P_{low}, which reduces the expansion ratio P_{high}/P_{low} for a given P_{high}. The optimiser compensates by increasing P_{high}, which explains the secondary upward trend along the T_{cf} axis. Conversely, at very low T_{cf} the condensation pressure is already low, so a moderate P_{high} is sufficient to achieve an adequate expansion ratio without incurring excessive pump work.

Constraint from the fixed power target. The 10 MW target couples P_{high} to the CO_2 mass flow rate. For a given P_{high}, the specific net work w_{net} is fixed by the cycle thermodynamics, and the mass flow adjusts to deliver $\dot{W}_{net} = \dot{m} \cdot w_{net} = 10\,\text{MW}$. This coupling means that any shift in P_{high} also changes the mass flow required, making the optimisation a true three-variable problem rather than a simple pressure maximisation.

Practical implications. The range of optimal pressures ($\sim$145 to 230 bar) sits at the upper end of the typical transcritical CO_2 operating window cited in Section 2.2.4 (80 to 200 bar). These values exceed those of conventional CO_2 refrigeration loops but remain below the 250 to 300 bar regime characteristic of purely supercritical Brayton cycles, confirming that the recuperated transcritical architecture occupies a moderate high-pressure niche.

3.5 Synthesis

Taken together, the four maps provide a consistent picture of the on-design behaviour of the recuperated transcritical CO_2 cycle. The net cycle efficiency improves monotonically with T_{hf} and with decreasing T_{cf}, reaching 20.3 % at the Carnot-favoured corner. This trend is consistent with the findings of Chen et al. [25] [11], who report a monotonic improvement of thermal efficiency with increasing heat-source temperature for recuperated transcritical CO_2 cycles. The absolute value of 20.3 % at $T_{hf} = 200\,^{\circ}\text{C}$ is higher than the 9–15 % range reported in comparable studies at similar source temperatures, which is consistent with the significantly colder cold sink used here ($T_{cf} = 0.1\,^{\circ}\text{C}$ versus the $\sim 20\,^{\circ}\text{C}$ typically assumed in waste-heat recovery contexts) and the higher source temperatures explored (130–200 °C versus 60–100 °C

in Wolf et al.[2]): both effects act cumulatively to increase the expansion ratio and the available enthalpy drop, as described in Section 3.4.

The existence of an optimal high-side pressure P_{high} at each operating point is a well-established result in the transcritical CO_2 literature. Guo et al. [22] explicitly confirm that high-side pressure is the key variable governing both efficiency and net power output of transcritical CO_2 Rankine cycles, and that a clear optimum exists. The monotonic increase of optimal P_{high} with T_{hf} found here (from $\sim$145 bar to $\sim$230 bar) is also consistent with the pressure levels reported by Wu et al. [20] for geothermal applications and by Yu et al. [27] for large-scale transcritical pumped thermal storage, where optimal pressures in the 100–200 bar range are identified at comparable source temperatures.

The CO_2 mass flow rate variation by a factor of approximately three across the explored envelope is consistent with the fixed-power scaling argument of Equation 21 and aligns with the order of magnitude reported by Cayer et al. [11], who found a CO_2 flow rate of 314.5 kg/s for a cycle of comparable specific work at a 100 °C source. The 4.7× variation in hot-source mass flow rate $\dot{m}_{hf}$ is consistent with the strong sensitivity to source temperature highlighted at MW-class scale by Manente et al. [26].

From the control-architecture perspective developed in Section 2.4, these maps will serve as reference trajectories for the supervisory controller of Chapter 4. The CO_2 mass flow rate is naturally regulated through the pump speed N_{pp}, the hot-source mass flow rate $\dot{m}_{hf}$ acts as a complementary actuator for the heat input, and the high-side pressure P_{high} emerges as the central controlled variable, a conclusion that is independently supported by the control-oriented analyses of [22, 8]. The boundary-temperature envelope explored here also defines the design space within which the off-design analysis must remain.

4 Off-Design Operation Analysis

4.1 Off-design problem definition and approach

Cycles designed for a fixed and constant set of operating conditions (the design point) can be optimised to achieve very high efficiency at that specific point. However, some real energy systems, such as Carnot batteries, rarely operate exactly at their design point: this is known as operating at an off-design point. Heat sources and sinks change with weather and seasons, electricity demand and supply vary over different patterns, and equipment performance degrades as it ages. The purpose of a robust control system is to continuously adjust the operating parameters of the cycle so that it performs as well as possible across the full range of real operating conditions. To reach this, the control system must know the current state of the cycle (temperatures, pressures, flow rates), have a model to predict how the system will respond to changes, and then make decisions.

Whereas the on-design study of Chapter 3 answered the question *"what is the best operating point for a given pair of source temperatures?"*, the off-design study answers a complementary question: *"given a cycle whose hardware is now fixed, how does it behave when the actuators and the boundary conditions move away from their nominal values?"* The answer to this question is what justifies and shapes the control architecture proposed at the end of this chapter.

The key conceptual difference with respect to Chapter 3 is that the component performance is no longer an input. In the design model, each component was characterised by key performance parameters (isentropic efficiency, effectiveness, pinch-point) listed in Table 1 and the geometry was never specified. For the off-design analysis this is reversed: the hardware is fixed at the geometry resulting from the design point, and the component performance becomes an output. The model is again built in Python with the *LaboThApPy* library, using *CoolProp* for the CO_2 equation of state.

The five components of the recuperated cycle are modelled as summarised in Table 3.

Component	**Type**	**Modelling approach**
Pump	Radial	Similarity-law extrapolation map
Expander	Axial	Meanline off-design model interpolated to a map
Recuperator	PCHE	Moving boundary (MB)
Gas heater	S&T	Moving boundary (MB)
Condenser	S&T	Moving boundary (MB)

Table 3: Component models used in the off-design study.

Pump (radial, similarity-law map). The CO_2 pump is described by a set of characteristic curves measured at a rated rotational speed. Operation at any other speed is obtained by extrapolating these curves through the pump similarity laws, which relate volumetric flow, head and power to the rotational-speed ratio.

Expander (axial, meanline interpolated to a map). The expander is an axial turbine described with a mean-line Aungier loss model. Because evaluating the full mean-line model at every cycle iteration would be computationally prohibitive, the model is run offline over a grid of inlet pressure and rotational speed to build a performance map, which is stored and then interpolated during the cycle solution. The map therefore returns the turbine mass flow and efficiency as a function of the inlet conditions and the expander speed N_{exp}. A relevant consequence of this modelling choice is that the expander geometry is fixed: there are no expander bypass valves.

Heat exchanger technology selection. The choice of heat exchanger type for each component is driven by the high operating pressures of the transcritical CO_2 cycle. On the high-pressure side, the working fluid can circulate at pressures between 145 and 230 bar (Figure 15), which places structural demands on the pressure capability of the equipment. Shell-and-tube (S&T) heat exchangers are selected for the gas heater and the condenser because the CO_2 can be routed through the tube side, where the small tube diameter keeps the hoop stress $\sigma = P\,D/(2\,t)$ low even at high pressures and allows the pressure boundary to be contained with reasonable wall thicknesses. The shell, which only sees the moderate pressure of the water-based hot or cold stream, can remain with a relatively thin large-diameter vessel without much structural penalty. In contrast, the recuperator operates as a gas-to-gas exchanger at moderate temperature differences with CO_2 on both sides. In this regime, a printed-circuit heat exchanger (PCHE) is preferred, as its diffusion-bonded construction combines compactness and pressure integrity without the large shell volume that would be required to contain high-pressure CO_2 on both streams in a S&T geometry.

It should nonetheless be kept in mind that operation above ~ 200 bar, although technically feasible, imposes non-negligible mechanical constraints. The components must be rated to sustain these pressures under daily cycling operation, which increases wall thicknesses and material requirements and therefore raises the capital cost (CAPEX) of the power block. The pressure level selected for operation is thus not only a thermodynamic choice but also an economic one.

Off-design closure: subcooling target and inventory. Once the geometry is fixed, neither the high-side pressure P_{high} nor the low-side pressure P_{low} can be set as inputs: both result from the balance between the pump, the expander and the heat exchangers. A closure condition

on the low-pressure side is however needed to keep the cycle well-defined. This is achieved by assuming a liquid CO_2 reservoir downstream of the condenser, which allows the condensation pressure to adjust so that the fluid reaching the pump is always slightly subcooled. In real plants such a receiver keeps suction conditions stable near saturation, but in this model the subcooling target is simply a mathematical tool to close the system of equations, avoiding the need to explicitly track the total fluid mass and the internal volume of the loop.

As discussed in Section 2.2.4, the fluid inventory directly controls the high-side pressure in transcritical CO_2 cycles [8, 22], and fixing the subcooling is equivalent to assuming that the liquid receiver is always full. A target of $SC = 3\,\mathrm{K}$ is used throughout the study, which gave stable convergence across the full operating range and is in line with similar transcritical CO_2 Rankine cycle studies [22]. Explicit mass conservation is left for future work (Section 5).

4.2 Off-design study methodology

The off-design behaviour is explored by varying the two turbomachinery speeds, the pump speed N_{pp} and the expander speed N_{exp}, which are treated as the active manipulated variables. They are swept on a two-dimensional grid spanning $\pm 15\,\%$ around their nominal values, while all geometric and source parameters are held fixed within each run. For every node of the grid, the cycle is solved, and the primary performance indicators,the net cycle efficiency η and the net electrical power $\dot{W}_{net}$, are recorded alongside the complete thermodynamic state of the cycle.

Once the model solves the complete thermodynamic state of the cycle, all other operational variables such as the high-side pressure P_{high} and the CO_2 mass flow rate $\dot{m}$, are directly obtained. Consequently, the high-pressure map is straightforwardly extracted from these simulation outputs. Furthermore, storing this comprehensive thermodynamic data provides a valuable diagnostic tool for future analysis, making it easier to resolve any underlying questions regarding the trends observed in the maps. This mapping process is then repeated for several boundary-condition cases, in order to assess the influence of source conditions on the actuator response.

It should also be noted that the off-design efficiencies of the electrical machines (the pump motor and the expander generator) are not modelled. A constant conversion efficiency is applied at every operating point, so the reported net electrical power and net efficiency capture the thermodynamic and turbomachinery behaviour only, and do not account for any part-load penalty of the electrical drives. This is a reasonable simplification given the relatively narrow speed range ($\pm 15\,\%$) explored here, but it should be relaxed if a wider operating envelope is considered.

The general structure of the off-design model, with the fixed hardware, the imposed boundary conditions, the two manipulated speeds and the output maps, is summarised in Figure 16.

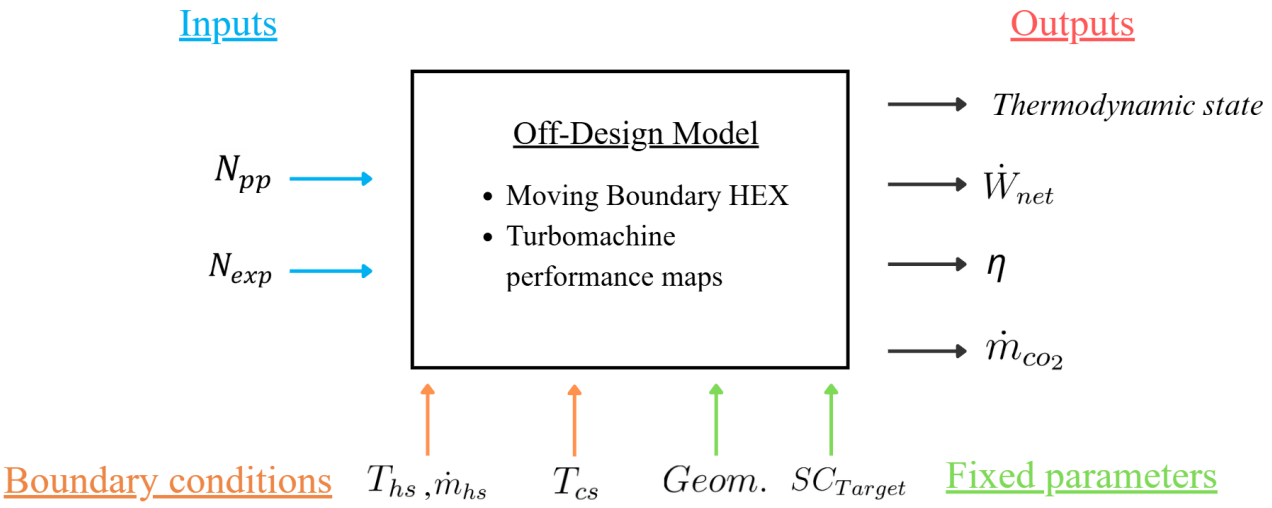


Figure 16: Off-design model structure: fixed component geometry and source conditions as boundary inputs, the pump and expander speeds (N_{pp}, N_{exp}) as manipulated variables, and the net efficiency, net power and the thermodynamic state of the cycle as outputs.

4.3 Off-design results and analysis

The off-design results are organised in four blocks. The first characterises the behaviour at the nominal operating point in order to establish a reference. The next three blocks examine, one at a time, how the speed-map response is reshaped when each of the three boundary conditions of the cycle (T_{hf}, T_{cf}, $\dot{m}_{hf}$) is moved away from its nominal value. A comparable parametric study under the same three boundary conditions has been carried out by Wu et al. [20] for a recuperated transcritical CO_2 cycle under a sliding-pressure strategy.

4.3.1 Nominal-point map

The reference (nominal) point for the off-design study lists the boundary conditions and the actuator setpoints to which the cycle hardware was matched during the design process. It is summarised in Table 4. The boundary conditions were selected to represent a typical intermediate operating point within the on-design envelope explored in Chapter 3, with a hot-source temperature of 150 °C and a cold-source temperature of 5 °C corresponding to moderate but realistic conditions for a transcritical CO_2 Carnot battery application. The high-side pressure, the CO_2 mass flow rate and any other internal variable are not part of the nominal specification: they emerge from the model itself once the geometry and the inputs are imposed.

Parameter	Symbol	Value
Hot-source temperature	T_{hf}	150 °C
Hot-source pressure	P_{hf}	5 bar
Hot-source mass flow rate	$\dot{m}_{hf}$	192.24 kg/s
Cold-source temperature	T_{cf}	5 °C
Cold-source pressure	P_{cf}	5 bar
Cold-source mass flow rate	$\dot{m}_{cf}$	4542 kg/s
Pump rotational speed	N_{pp}	2900 rpm
Expander rotational speed	N_{exp}	2864.77 rpm
Subcooling target	SC	3 K

Table 4: Nominal operating point used as reference for the off-design study.

The reference (N_{exp}, N_{pp}) maps are obtained at the nominal boundary conditions and they serve three purposes: (i) they show the shape of the cycle-level output maps (η, $\dot{W}_{net}$, P_{high}, $\dot{m}$) and of the component-level maps (pump and expander isentropic efficiencies, η_{pump} and η_{exp}, and expander pressure ratio Π_{exp}) around the design point; (ii) they identify which of the two speeds has the strongest influence on each output; and (iii) they provide the baseline against which the boundary-condition sweeps of the following subsections are compared.

The cycle-level maps in Figure 17 reveal the response of the fixed-geometry cycle to the two manipulated speeds around the nominal point, while the component-level maps in Figure 18 make explicit the turbomachinery behaviour that underlies that response.

Pressure and mass flow are governed by the pump. The two lower maps in Figure 17 show nearly horizontal contour lines, which means that both the high-side pressure P_{high} and the CO_2 mass flow rate $\dot{m}$ are controlled almost exclusively by the pump speed N_{pp}, with very little influence from the expander speed N_{exp}. Increasing N_{pp} raises the mass flow rate, which in turn raises the high-side pressure. This confirms and extends the qualitative chain anticipated in Section 2.4,

$$N_{pp} \uparrow \Rightarrow \ \dot{m} \uparrow \Rightarrow \ P_{high} \uparrow, \tag{23}$$

and establishes N_{pp} as the natural actuator for regulating the operating pressure of the cycle, in line with its role as the primary flow-regulating actuator identified by Hernández et al. [23]. The expander pressure ratio Π_{exp} in Figure 18 mirrors this behaviour: its contour lines are also close to horizontal, so Π_{exp} rises with N_{pp} (more flow, higher upstream pressure) and is only weakly reduced by N_{exp}. The pressure ratio is thus a direct image of the pump-driven pressure build-up rather than an independent degree of freedom of the turbine.

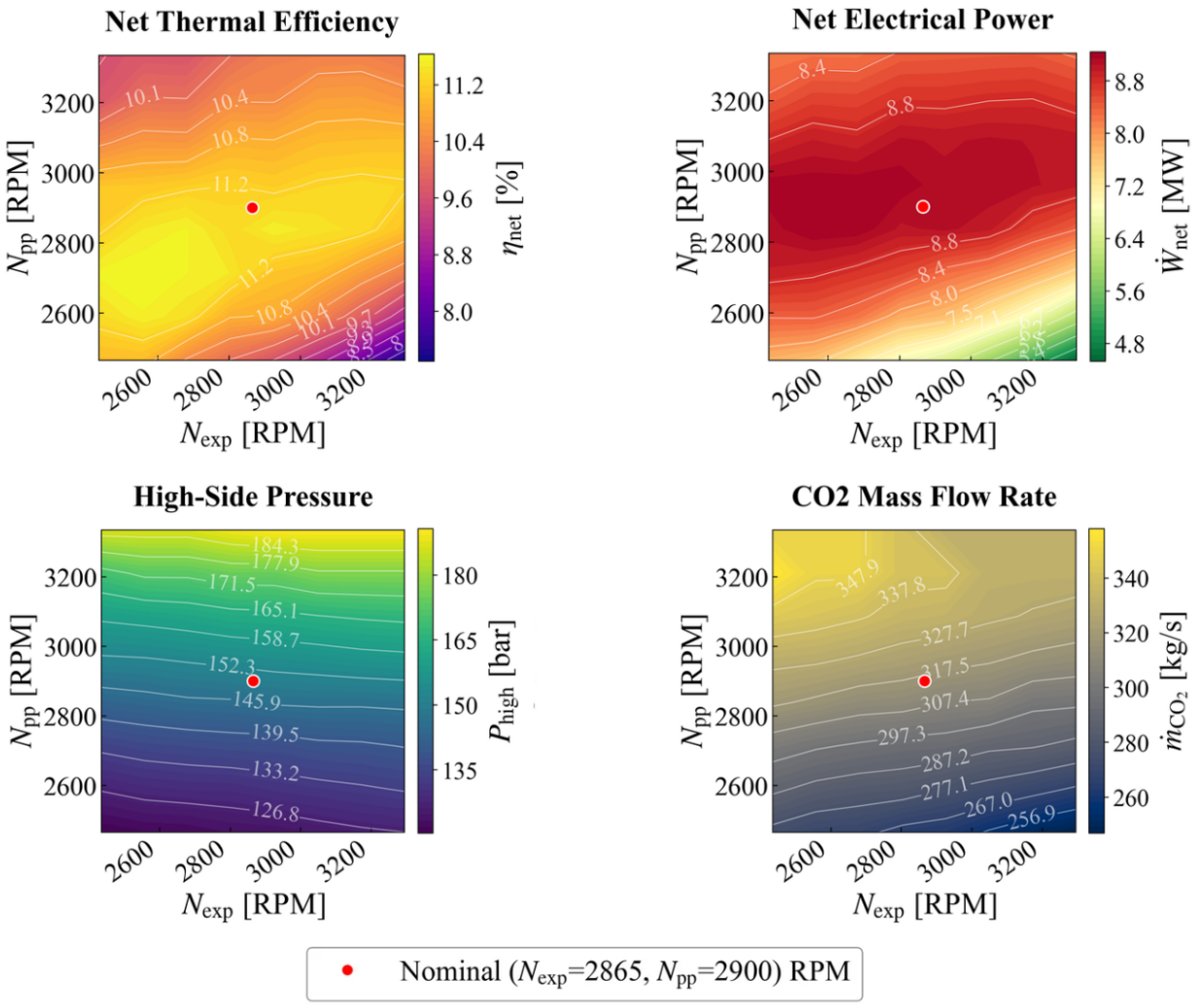


Figure 17: Cycle-level off-design performance maps at the nominal boundary conditions (Table 4) as a function of the expander and pump rotational speeds (N_{exp}, N_{pp})

The maximum efficiency and the maximum power are not in the same zone. Comparing the two upper maps in Figure 17, the peak of net thermal efficiency η sits at a slightly lower pump speed than the peak of net electrical power $\dot{W}_{net}$. The reason is simple: net power keeps growing as the mass flow increases, whereas efficiency is power divided by heat input, so it peaks at a smaller flow. The same offset has been reported for sCO_2 Brayton cycles [30]. Because the two optima are close but not identical, improving one slightly worsens the other.

The nominal point does not coincide with either optimum. The nominal operating point (red marker) does not exactly match either the maximum efficiency point or the maximum net power point. This is expected because the pump and expander were designed separately, each using its own design conditions. As a result, the nominal cycle operating point is not necessarily the optimum of the complete system. However, this difference is not significant. Around the nominal point, both the cycle efficiency and the net power output vary only slightly with pump speed. Therefore, small deviations from the nominal speed have a limited impact on overall performance, making the operating point robust. In addition, the component maps show that both turbomachines operate close to their intended design conditions. The pump works near its highest-efficiency region, while the expander remains within a broad high-efficiency zone. Consequently, the nominal operating point provides a good compromise between efficiency, power output, and stable operation.

The expander becomes disadvantageous at low pump speed. In the lower-right corner (N_{exp} high, N_{pp} low), both η and $\dot{W}_{net}$ collapse to their lowest values. The component maps explain why: η_{exp} drops from $\approx 88\,\%$ to below $65\,\%$ in exactly the same corner (Figure 18). A low pump speed delivers insufficient flow while a high expander speed increases the turbine swallowing capacity, pushing the blade loading away from its design condition and driving up incidence losses. The pressure ratio Π_{exp} also drops simultaneously, so a less efficient expansion across a smaller pressure ratio produces far less work. Raising N_{exp} is therefore only beneficial when N_{pp} already sustains the flow.

The pump is the dominant actuator near the nominal point. Inspecting the maps visually along each axis around the nominal setpoint reveals a clear asymmetry: varying N_{pp} alone produces large gradients in both η and $\dot{W}_{net}$, while varying N_{exp} alone at the same N_{pp} causes comparatively little change. Near the nominal point, therefore, N_{pp} holds most of the control authority over cycle outputs. This dominance does not hold across the full map: as shown in the previous paragraph, at low N_{pp} the expander speed regains strong (and adverse) influence through the collapse of η_{exp}.

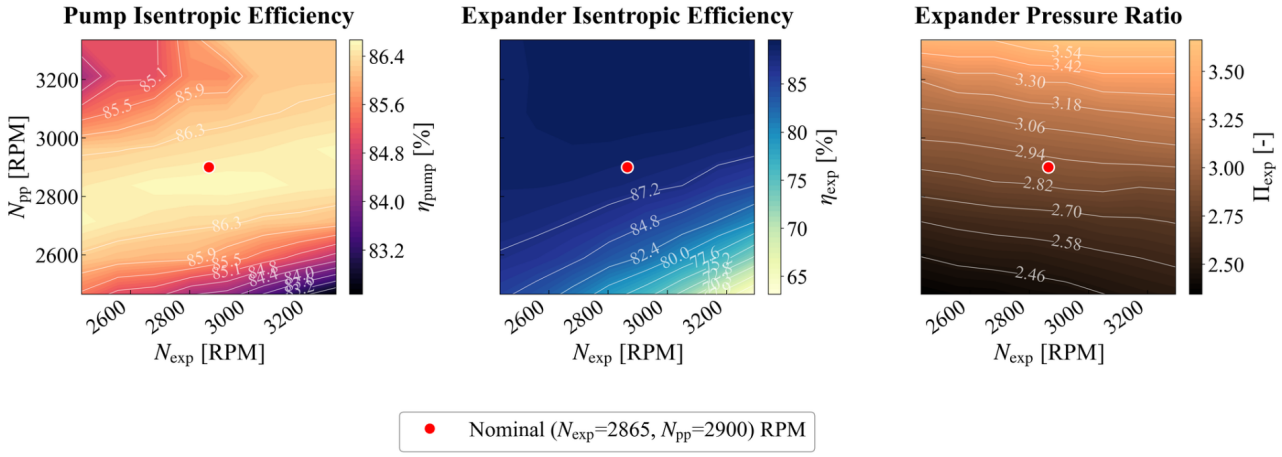


Figure 18: Component-level off-design maps at the nominal boundary conditions: pump isentropic efficiency η_{pump}, expander isentropic efficiency η_{exp}, and expander pressure ratio Π_{exp}, as a function of the expander and pump rotational speeds (N_{exp}, N_{pp}).

4.3.2 Sensitivity to the hot-source temperature (T_{hf} sweep)

The (N_{exp}, N_{pp}) map is regenerated for five values of the hot-source temperature (135, 145, 160, 175 and 190 °C), keeping T_{cf} and $\dot{m}_{hf}$ at their nominal values. The results are stacked in Figure 19, with a common colour scale to allow direct visual comparison.

Efficiency and power increase with T_{hf}. A higher source temperature delivers more enthalpy to the CO_2 in the gas heater, increasing the enthalpy drop across the expander. At the nominal actuator setpoint, η rises from 9.2 % at 135 °C to 14.6 % at 190 °C, and $\dot{W}_{net}$ from 7.1 to 13.1 MW, with no change in actuator setpoints or hardware. Importantly, the shape of the maps is preserved across all five rows: T_{hf} shifts the performance level uniformly without altering the actuator–output relationships established at the nominal point. This rise of η and $\dot{W}_{net}$ with T_{hf} is the off-design counterpart of the trend already found in the on-design study of Section 3.4, where the optimised net efficiency increased monotonically with the hot-source temperature across the whole envelope.

High-side pressure and mass flow are unaffected by T_{hf}. The pressure and mass-flow maps are visually identical across all five rows: P_{high} stays within 152.7–153.8 bar and $\dot{m}$ varies by less than 2 kg/s across the entire 55 °C temperature sweep. This is a direct consequence of the pump-dominated closure of Section 4.1: the hydraulic state of the cycle is set by N_{pp} alone and is insensitive to the hot-source boundary condition. From a control perspective this decoupling is convenient, since pressure and flow can be regulated through N_{pp} independently of any fluctuation in the heat source.

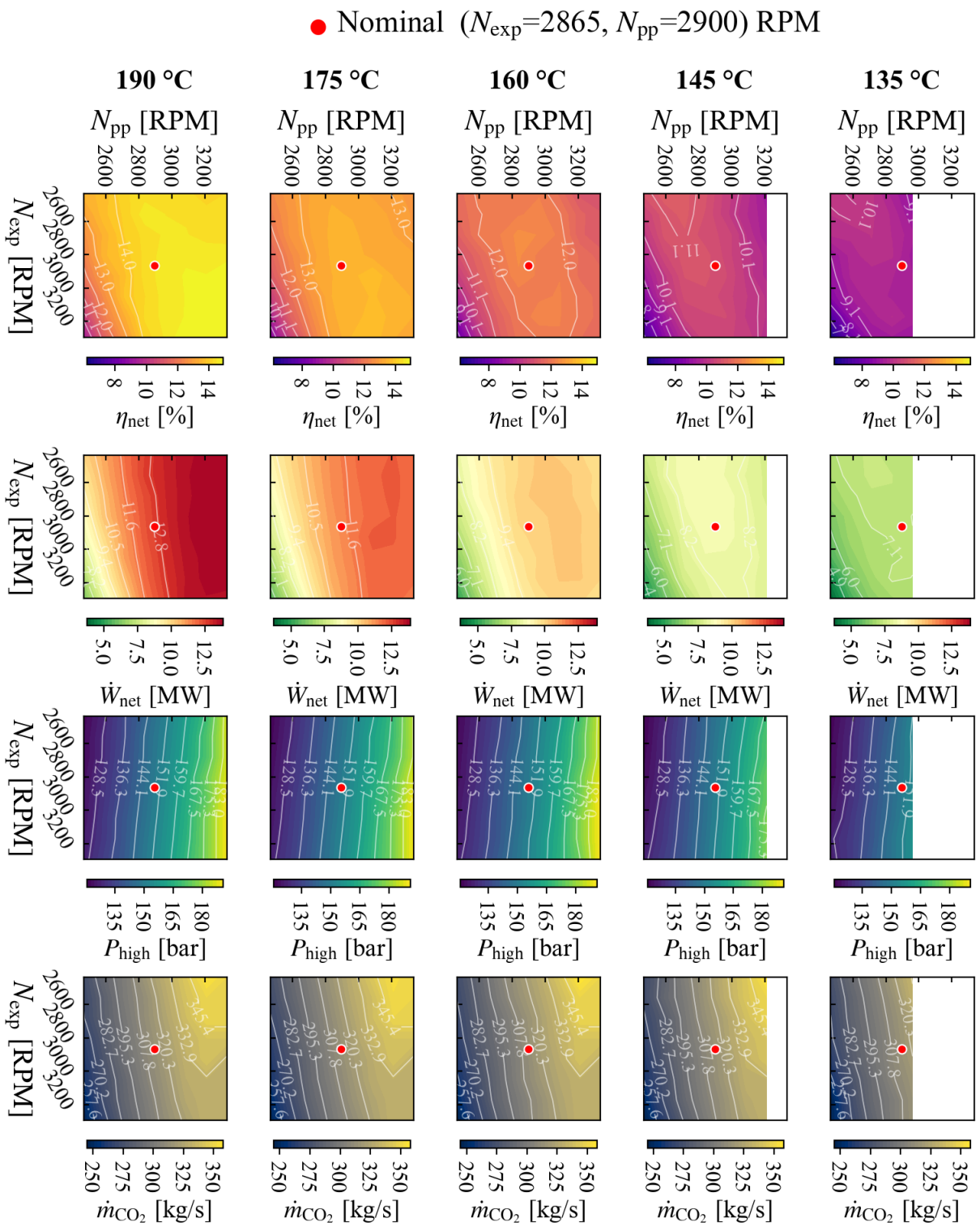


Figure 19: Off-design performance maps for five values of T_{hf}. Each row shares a common colour scale with the nominal-point maps of Figure 17.

Convergence is lost at low T_{hf} combined with high N_{pp}. One region of the map could not be solved: when a low hot-source temperature is combined with a high pump speed, the model fails to converge and the corresponding nodes are left blank in Figure 19. In this corner the pump forces a high mass flow into a gas heater that, at low T_{hf}, can supply only a limited amount of heat, so the CO_2 leaves the heater relatively cold and the low-pressure iteration struggles to satisfy the prescribed subcooling. Two interpretations are possible. The failure may be purely numerical, since the solver is already sensitive in this region and requires a damping strategy and several pressure initial guesses to converge elsewhere. Alternatively, it may reflect a genuine physical limit, in which case real operation at low T_{hf} and high N_{pp} would also become unstable. A practical way to distinguish the two, and a useful operating guideline, is to relax the subcooling target: increasing SC enlarges the liquid margin at the pump inlet and is expected to restore convergence over part of this corner. The subcooling target should therefore be regarded as a genuine operating constraint rather than an arbitrary closure value, a point revisited in the control guidelines of Section 4.4

4.3.3 Sensitivity to the cold-source temperature (T_{cf} sweep)

The same exercise is repeated for five values of the cold-source temperature T_{cf}, with the hot source held at its nominal conditions (150 °C). The cycle-level maps are shown in Figure 20 and the component-level maps (pump and expander efficiencies and expander pressure ratio) in Figure 21. The conclusions are the following.

Efficiency and power fall as T_{cf} rises. A warmer sink raises the condensation temperature, and with it the low-side pressure P_{low}. This reduces the expander pressure ratio $\Pi_{exp} = P_{high}/P_{low}$ (from 2.6–3.9 at 0.1 °C to 1.9–2.5 at 20 °C as shown in Figure 21) and the enthalpy drop available to the expander, so the cycle produces less work per kilogram of CO_2. At the nominal setpoint, raising T_{cf} from 0.1 to 20 °C lowers η from 11.5 % to 9.0 % and nearly halves the net power, from 10.1 to 5.4 MW. This is the same Carnot trend already seen in the on-design map of Section 3.4.

The mass flow now responds to T_{cf}. A clear difference with respect to the T_{hf} sweep is worth highlighting. The high-side pressure remains nearly constant (153.8–155.0 bar), confirming that P_{high} is governed by N_{pp} rather than boundary conditions. The CO_2 mass flow decreases from 330 to 291 kg/s as T_{cf} rises. This effect originates at the pump inlet: a warmer cold source increases the suction temperature, pushing the fluid closer to the critical point and reducing its density from 883 to 735 kg/m^3. Because the pump runs at fixed speed, the volumetric flow changes only slightly (0.374 → 0.396 m^3/s), so $\dot{m} = \rho \dot{V}$ leads to a lower mass flow. This reduction, together with the lower specific work from a reduced pressure ratio, drives the drop in net power. The rise in low-side pressure (46 → 70 bar) also confirms the relation $T_{cond} \uparrow \Rightarrow P_{low} \uparrow$ from Section 2.4.

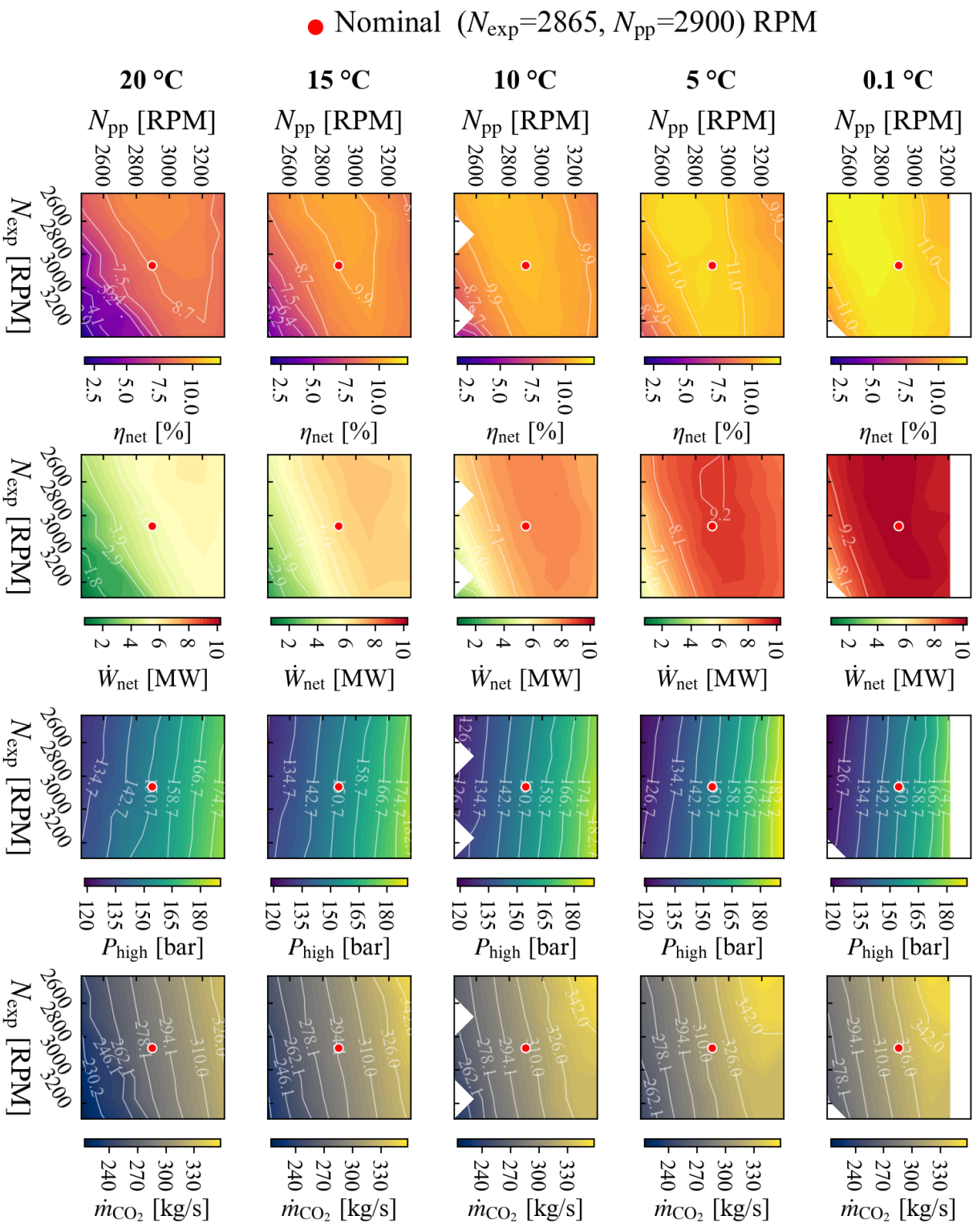


Figure 20: Cycle-level off-design performance maps for five values of T_{cf}.

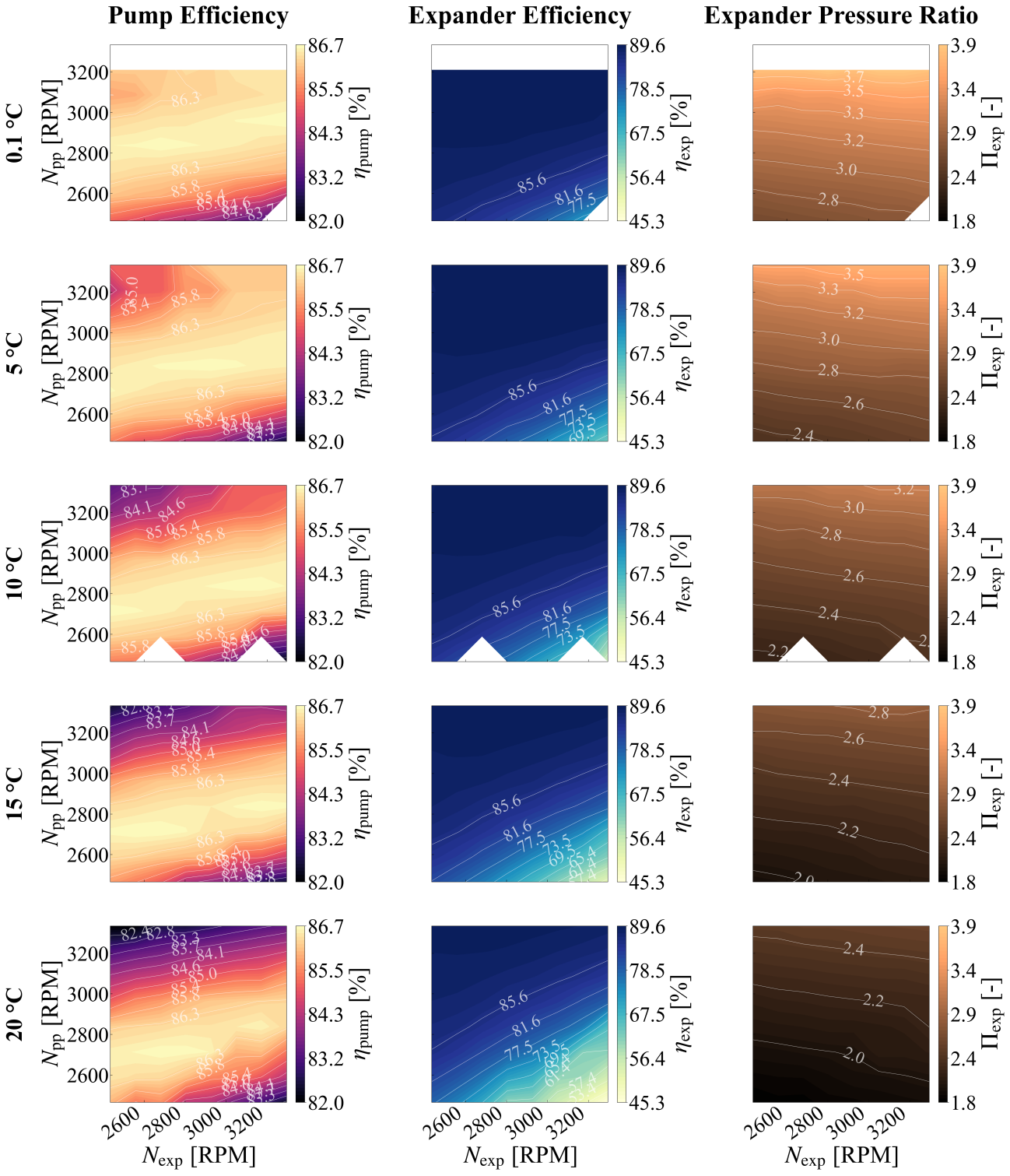


Figure 21: Component-level off-design maps for five values of T_{cf}: pump efficiency η_{pump}, expander efficiency η_{exp} and expander pressure ratio Π_{exp}.

A warmer sink also penalises the expander efficiency. The expander-efficiency maps in Figure 21 reinforce the same picture. The high-efficiency plateau (top-left) stays near 89 % in every case, but the low-efficiency corner (bottom-right) deepens markedly as T_{cf} rises: the minimum η_{exp} falls from about 72 % at 0.1 °C to roughly 45 % at 20 °C, and the poor-efficiency region grows larger. The reason is the same kinematic mismatch described at the nominal point: as Π_{exp} shrinks, a given expander speed is increasingly oversized for the reduced pressure ratio and the losses grow. A warmer sink thus hurts the cycle twice over: it lowers the pressure ratio and the specific work, and it pushes the expander into a less efficient operating regime. A direct corollary is that the expander speed becomes a more influential control variable as T_{low} rises: whereas N_{exp} played only a fine-tuning role at the nominal point, at a warm sink its correct selection is needed to keep the expander out of the deepening low-efficiency corner.

The pump high-efficiency band narrows as T_{cf} rises. At low T_{cf} the pump operates on liquid well below the critical point, where the density is high and stable; a wide range of speeds keeps the pump near its best-efficiency flow, so the high-η_{pump} band is broad. As T_{cf} rises, the suction state moves closer to the critical point, where small changes in operating conditions cause large swings in density. The pump then matches its best-efficiency point only over a narrower window of speeds, and the high-efficiency band shrinks accordingly.

4.3.4 Sensitivity to the hot-source mass flow rate ($\dot{m}_{hf}$ sweep)

The third boundary condition examined is the hot-source mass flow rate $\dot{m}_{hf}$, swept symmetrically around its nominal value (Table 4) while both source temperatures are held constant. The maps are shown in Figure 22.

Diminishing returns of $\dot{m}_{hf}$ on efficiency and power. Increasing the hot-source mass flow rate raises both the net efficiency and the net power, consistent with the analysis presented in Section 2.4, but with a clearly decreasing return, as shown in the graph in Figure 23. The first increments are the most effective: raising $\dot{m}_{hf}$ from 154 to 173 kg/s adds 1.3 MW, whereas the last increment from 211 to 231 kg/s adds only 0.1 MW, and the efficiency levels off near 12 %. This saturation is consistent with the heat-transfer limit of the gas heater: once the hot-fluid flow is large enough to supply the heat the CO_2 side can absorb, further increases mainly reduce the hot-fluid temperature glide rather than transferring additional heat to the cycle.

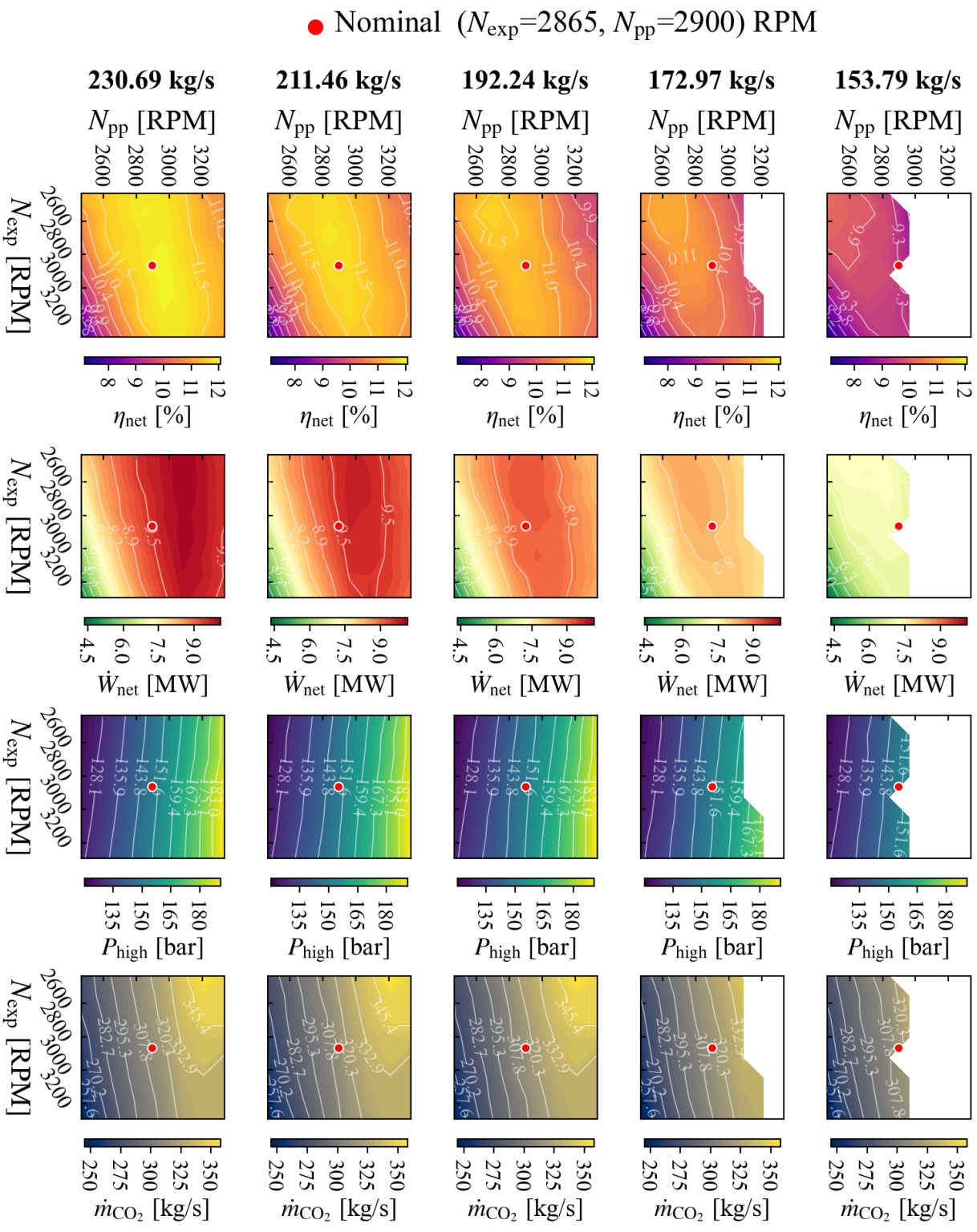


Figure 22: Off-design performance maps for five values of $\dot{m}_{hf}$.

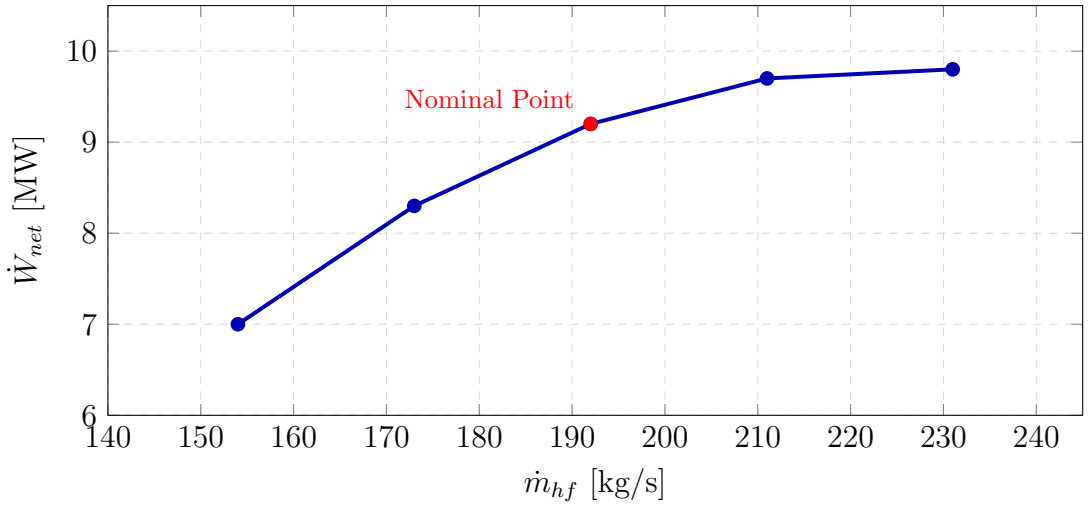


Figure 23: Net power as a function of the hot-source mass flow rate $\dot{m}_{hf}$ at nominal actuator setpoints ($N_{pp} = 2900$, $N_{exp} = 2865$) rpm. The curve highlights the diminishing returns and the power ceiling effect beyond the nominal operating point.

More hot-source flow widens the pinch and unlocks higher pump authority. As shown in Figure 24, a low $\dot{m}_{hf}$ steepens the hot-source profile, tightening the pinch and limiting the heat the gas heater can transfer regardless of pump speed. A higher $\dot{m}_{hf}$ flattens the red curve, opens the temperature gap, and removes this thermal ceiling. The consequence is direct: at the nominal flow the net power saturates at 9.2 MW, while at $\dot{m}_{hf} = 230.7$ kg/s raising N_{pp} to 3086 rpm delivers 10.1 MW. In short, $\dot{m}_{hf}$ sets the thermal ceiling within which N_{pp} operates: without sufficient flow from the hot source, increasing the pump speed offers little gain.

High-side pressure and CO_2 mass flow rate remain insensitive to $\dot{m}_{hf}$. The P_{high} and $\dot{m}$ maps are again essentially identical across the five cases. At the nominal setpoint, P_{high} stays within 153.3–154.5 bar and $\dot{m}$ within 319–326 kg/s despite a 50 % change in $\dot{m}_{hf}$. The pump-suction diagnostics confirm why: the liquid state at the pump inlet barely moves (density ≈ 850 kg/m^3, suction temperature $\approx 12\,^\circ$C across all cases), exactly as in the T_{hf} sweep. As with the temperature sweep, the hot-source flow rate is not a hydraulic actuator on the cycle: it modulates only the energy input. Neither hot-source boundary condition therefore acts on the pump–expander equilibrium; that role belongs to T_{cf} on the boundary side and to the turbomachinery speeds on the manipulated side.

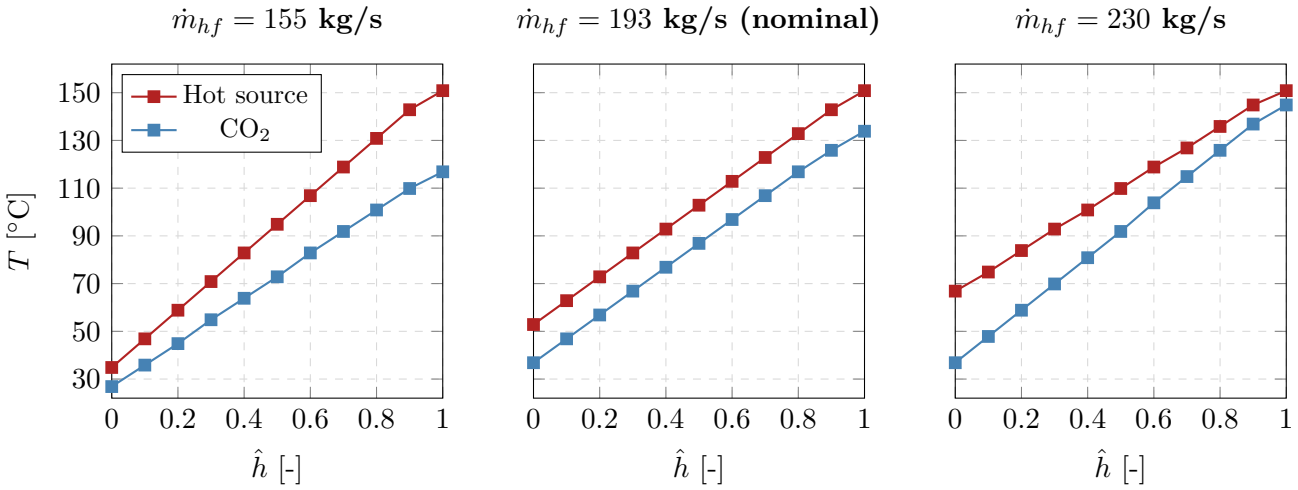


Figure 24: Gas heater temperature profiles for three hot-source mass flow rates. Increasing $\dot{m}_{hf}$ flattens the hot-source (red) curve and widens the pinch point, keeping the heat exchange feasible; the nominal case sits in between.

4.4 Control architecture guidelines

The off-design study has mapped how the cycle responds to its two manipulated speeds (N_{pp}, N_{exp}) and to its three boundary conditions ($T_{hf}, T_{cf}, \dot{m}_{hf}$). The observations of the preceding subsections can be condensed into a small set of guidelines that should inform any future control architecture for this power block. They are organised below by the variable that drives each effect.

Across every sweep, the high-side pressure P_{high} and the CO_2 mass flow rate $\dot{m}_{CO_2}$ were governed almost exclusively by N_{pp} and were essentially insensitive to the thermal boundary conditions. The pump speed should therefore be the fast actuator dedicated to regulating the hydraulic state of the cycle, while the expander speed N_{exp} retains only a fine-tuning role near the nominal point. This pump-driven, sliding-pressure operation, in which P_{high} floats to the equilibrium set by the turbomachinery rather than being throttled, is the strategy consistently adopted in off-design studies of transcritical CO_2 cycles [20, 31, 32].

The expander speed must follow the pump rather than lead it. The nominal map showed that raising N_{exp} is beneficial only when N_{pp} already sustains enough flow; otherwise the expander runs oversized for the available pressure ratio, its efficiency collapses, and both η and $\dot{W}_{net}$ fall. A practical controller should therefore coordinate the two speeds, keeping the expander matched to the flow delivered by the pump rather than letting it run ahead. In practice, since N_{exp} plays only a fine-tuning role near the nominal point, it may even be held constant over a moderate operating window, adjusting it only when the sink temperature rises enough to push

the expander towards its low-efficiency corner.

A change in the hot-source temperature T_{hf} shifts the efficiency and power maps up or down uniformly without altering the actuator–output relationships. This is convenient for control: pressure and flow can be regulated through N_{pp} independently of any fluctuation in the heat source. The only condition is the loss of convergence observed at low T_{hf} and high N_{pp}, which signals that the subcooling target SC is a genuine operating constraint and may need to be raised to keep the feasible actuator space open under cold-source-limited conditions.

Unlike the two hot-source boundaries, the cold-source temperature T_{cf} acts directly on the pump-suction state: a warmer sink lowers the liquid density and reduces the mass flow at fixed pump speed, while also decreasing the expander pressure ratio and pushing the expander into a less efficient regime. The key control implication is that a warmer sink requires an increase in N_{pp} to compensate for the density loss and maintain the design mass flow rate. The controller should therefore treat T_{cf} as the principal measured disturbance and feed it forward to the pump-speed loop. The benefit of a sliding-pressure operation adapted to the sink temperature is confirmed by Du et al. [31] in a transcritical CO_2 cycle.

The hot-source mass flow rate $\dot{m}_{hf}$ behaves as an energy regulator with diminishing returns. Increasing it raises the heat available to the cycle and, combined with a matching increase in N_{pp}, allows more heat to be recovered and more power to be produced. However, the benefit saturates once the gas heater reaches its heat-transfer limit: beyond that point, extra hot-source flow mainly reduces the temperature glide rather than adding useful heat. From an operational standpoint there is therefore no need to circulate the maximum hot-source flow; $\dot{m}_{hf}$ should be modulated only up to the point where its marginal power gain justifies the additional auxiliary pumping cost. This is further reinforced by the Carnot-battery context: since the hot source is a stored thermal reservoir, excessive hot-source flow leaves residual heat at the exhaust of the heat-source circuit that is not transferred to the cycle, wasting stored energy that could otherwise contribute to power generation.

Taken together, these guidelines define a clear variable structure for the power block: P_{high} emerges as the central controlled variable, regulated through N_{pp}; T_{cf} is the main disturbance to tackle; and $\dot{m}_{hf}$ and N_{exp} act as secondary, slower actuators for adjusting the heat input and the expander match. This ordering is consistent with the transcritical CO_2 control literature, in which the high-side pressure is repeatedly identified as the dominant variable on both efficiency and net power [22, 8].

5 Conclusions

This thesis has presented a control-oriented thermodynamic analysis of a recuperated transcritical CO_2 Rankine cycle intended for the discharge stage of a Carnot battery. Starting from the manipulated and controlled variables identified in the literature review, the study quantified their relative importance through two complementary models: an on-design optimisation study and an off-design, fixed-geometry study, both implemented in Python with the LaboThApPy library and CoolProp. The main outcomes can be summarised as follows.

On-design performance mapping. A Particle Swarm Optimisation was used to locate, for every pair of source temperatures, the operating point maximising net cycle efficiency at a fixed net power of 10 MW. The resulting performance maps showed a monotonic improvement of efficiency with increasing hot-source temperature and decreasing cold-source temperature, reaching a global optimum of 20.3 % (about 48 % of the corresponding Carnot limit) at the Carnot-favoured corner, with an optimal high-side pressure of 218 bar. Across the explored envelope the optimal CO_2 mass flow rate varied by a factor of about three and the hot-source mass flow rate by a factor of about four, confirming the strong sensitivity of the cycle to its boundary conditions. These maps establish reference set-points and demonstrate that an optimal high-side pressure exists at every operating point, in agreement with the transcritical CO_2 literature [11, 25].

Off-design behaviour and actuator authority. By keeping the hardware fixed and reversing the role of the component models (performance as an output rather than an input) the off-design study clarified how the cycle responds to its two manipulated speeds and to its boundary conditions. The high-side pressure and the CO_2 mass flow rate proved to be governed almost exclusively by the pump speed N_{pp} and essentially decoupled from the thermal boundary conditions. The expander speed N_{exp} retained only a fine-tuning role near the nominal point and became unfavourable at low pump speed, where the turbine runs oversized for the available pressure ratio and its efficiency collapses. Among the boundary conditions, only the cold-source temperature acted on the hydraulic state of the cycle, by altering the pump-suction density; the hot-source temperature and mass flow rate modulated only the thermal level of the maps, with the hot-source flow exhibiting clearly diminishing returns once the gas-heater heat-transfer limit was reached.

Proposed control architecture. Taken together, the two studies define a clear variable structure for the power block. The high-side pressure emerges as the central controlled variable and should be regulated through the pump speed. The cold-source temperature should be treated as the principal measured disturbance and fed forward to the pump-speed loop. The hot-source mass flow rate and the expander speed act as secondary, slower actuators for adjusting the heat input and keeping the expander matched to the flow delivered by the pump. This

ordering is consistent with the conclusions of the transcritical CO_2 control literature, in which the high-side pressure is repeatedly identified as the dominant variable for both efficiency and net power [8, 22].

Limitations. The analysis is entirely steady-state: it characterises the family of feasible equilibrium operating points and the static sensitivities around them, but it does not capture transient dynamics, thermal inertia or closed-loop stability. No controller has yet been implemented or tuned, and no experimental validation has been carried out; the results rest on component models calibrated with effectiveness, efficiency and pinch-point values taken from the literature. A single cycle architecture (the recuperated layout) and a single working-fluid charge have been considered, and the working-fluid inventory (known to govern the high-side pressure [22]) has been treated only implicitly through the sub-cooling closure rather than as an explicit manipulated variable.

Future work. The next step is the development of a dynamic model of the power block [3] so that the proposed variable structure can be tested in closed loop, comparing conventional PID regulation with non-linear control methods such as model -predictive control of the high-side pressure under realistic disturbance scenarios [23, 33]. The explicit modelling of charge/inventory management, the extension of the study to alternative architectures and the coupling of the power block with the heat-pump (charge) stage to evaluate the full round-trip efficiency [4, 26] are also of clear interest. Experimental validation on a laboratory-scale rig [34] and a techno-economic assessment of the resulting control strategies would complete the path from the present analysis towards a deployable transcritical CO_2 Carnot battery.

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